

Coherent Full-Duplex Double-Sided Two-Way Ranging and Velocity Measurement Between Separate Incoherent Radio Units

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Abstract—This paper presents a novel coherent full-duplex (FD) double-sided two-way ranging (CFDDS-TWR) technique for wireless locating and velocity measurement between separate radio units. The idea presented here is a quantum leap in wireless locating since it enables coherent ranging, continuous phase tracking, and velocity measurement between wireless units that operate incoherently with separate low-cost crystal oscillator clock sources. Frequency modulated continuous wave chirp sequences are exchanged in a FD manner between two radio units. The signals transmitted from two radio units are received, respectively, by each of the two units and down-converted in a mixer with the receiver's own transmit signal to generate a beat signal in each unit. The two beat signals are then processed together, after one beat signal is transmitted to the partner unit. Phase coherent range and Doppler phase measurements can be conducted between incoherent radio units. The presented CFDDS-TWR technique also offers options for reducing noise-like distortions caused by mixing the products of uncorrelated phase noise. We show both analytically and experimentally that the described compensation method reduces the noise level and enhances the dynamic range of the ranging signals tremendously—as evidenced by an improvement of around 24 dB in our results. These unique signal and synchronization properties show that the CFDDS-TWR method is a highly accurate measurement approach. Results acquired with a 24-GHz test system in a 7- to 17-m range demonstrate a standard deviation in range and velocity of 0.25 mm and 0.05 mm/s, respectively.

Index Terms—Doppler radar, full-duplex (FD), phase measurement, phase noise (PN), radar, radio navigation, synchronization, transponders, wireless localization.

I. INTRODUCTION

WIRELESS locating describes the estimation of the location of an active or a passive transceiver by means of the time delay due to the propagation time in a wireless channel [1]. A wide range of use cases for this technology

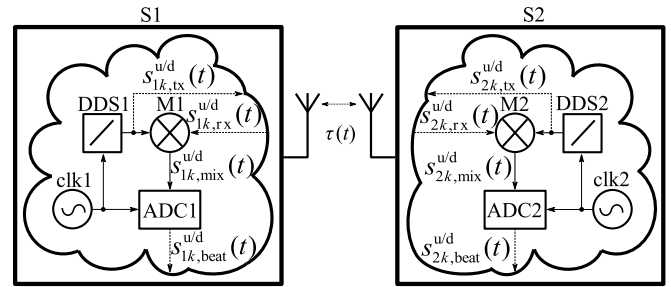


Fig. 1. Setup of two FMCW secondary radar stations (S1 and S2) with individual system clock sources (clk1 and clk2), RF generators (DDS1 and DDS2), frequency mixers (M1 and M2), and analog-to-digital converters (ADC1 and ADC2). The transmission delay $\tau(t)$ over the wireless channel is time dependent.

are to be found in manufacturing, logistics, the internet of things, and wireless consumer products [2]. Among many other techniques, the round-trip time-of-flight method, often called the two-way ranging (TWR) method, is a standard approach for wireless locating [3]. As shown in Fig. 1, the total time of a two-way communication between two radio units is determined. An advantage of the TWR method is that it does not require the correction of the timing offset. Only the clock rate in the two units involved must be synchronized to correctly account for the processing time if the response is not sent immediately by the responding unit [4].

Concerning the modulation format, the main advantage of modulated continuous wave (FMCW) systems is their economical implementation in terms of hardware requirements [5]. FMCW-based wireless locating by means of ToF and time difference of arrival is described in [6] and [7]. Ultra-wideband systems are also available: their superior range resolution makes them more robust in multipath environments [8]. Inaccurate positioning is feasible with a received signal strength (RSS) approach based on an existing infrastructure of wireless local area networks [9]. In this case, orthogonal frequency division multiplexing modulation is mostly chosen for wireless position estimation, as discussed in [10].

All methods mentioned above rely on incoherent clock sources and do not allow coherent radar measurements between separate units. Sophisticated phase synchronization is known from airborne and satellite bistatic synthetic aperture radar (SAR) systems [11]. Ultrahigh-quality oscillators are an option for practical use, but they are far too expensive

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for commercial positioning applications [12]. Exchanging frequent synchronization pulses to correct for the phase deviation is a viable workaround here [13]. This approach does not compensate for the impact of phase noise (PN) during sampling pulses as different signals with time division multiplexing are used for synchronization and sampling.

The approach presented in [14] and [15] describes a self-synchronizing bistatic radar technique with analog chaotic circuits to obtain information on range and Doppler shift of targets. Our method is different, as it does not detect passive targets but the transmission between two active radar units.

Multistatic FMCW radar systems based on over-the-air deramping (OTAD) are explained in [16]. With this technique, coherent signal processing becomes feasible and target tracking based on a moving target indicator has been demonstrated. The drawbacks are the second frequency band, the increased hardware and computational requirements, and the need for an LOS between each radar node and the master station.

Secondary radar systems suffer from PN, which impairs the accuracy of ToF measurements. But in contrast to primary radar, both stations' PN is uncorrelated in secondary radar. This limitation is overcome by the use of analog-to-digital conversion (ADC) data in both wireless units [15]. PN and nonlinearities are compensated inherently, and highly precise range measurements have been demonstrated in an operational region where PN has a more severe impact than additive white Gaussian noise (AWGN).

In [18], two wireless radar stations are physically connected with a ramp trigger that ensures accurate timing of individual FMCW chirps. In-chirp PN is suppressed, and range measurement accuracy is improved by multiplying both intermediate frequency (IF) signals in the time domain. This nonlinear operation leads to a spectral convolution and causes ghost targets in multipath channels. And the need for a physical connection restricts mobility in full wireless applications.

Velocity estimation with FMCW primary radar and consecutive up- and down-chirps via beat frequency estimation has been presented a long time ago [19]. Modern FMCW primary radar systems perform a fast Fourier transform (FFT) across the "slow-time" dimension (the phase shift of consecutive FMCW chirps is dealt with in this way) [20]. This technique is more accurate; it avoids incorrect target assignments; and is computationally efficient.

Our paper explains the coherent operation of separate wireless radar stations with incoherent low-cost oscillators. Computation and correction of unknown offsets from ADC data and the creation of a synthesized beat signal enable the suppression of systematic errors and PN. Hence, ranging accuracy will be enhanced and precise coherent velocity and displacement measurements become possible.

This paper is structured in the following order: the CFDDS-TWR measurement principle and the system setup will be introduced first. The presynchronization requirements are then defined. In Section III, a general signal model for CFDDS-TWR with two wireless radar units and coarse presynchronization will be derived. By simultaneous transmission and reception, unknown parameters will be estimated based on ADC data and the reconstruction of a synthetic beat signal

will be explained in Section IV. Range and velocity estimation and phase tracking will be described. In Section V, the CFDDS-TWR signal model with precise presynchronization by means of timing drift compensation will be derived and multiplexing with more than two radar units will be elaborated. Finally, simulation results of the synthetic beat signal in Section VI and measurement results in Section VII will be given to verify theoretical concepts with empirical data.

II. CFDDS-TWR SYSTEM SETUP

Initially, this section reviews the system setup with two conventional FMCW radar units and defines relevant signals. Thereafter, the CFDDS-TWR measurement principle is introduced and assumptions for the required presynchronization quality will be given.

A. System Setup and Signal Definitions

The radar units (the words "stations" and "units" are used synonymously) are depicted in Fig. 1 and labeled as station 1 (S1) and station 2 (S2). Both stations are entirely wireless and rely on individual local oscillators (LOs) as system clock sources (clk1 and clk2). Therefore, there is a relative drift in time δ_t , a Doppler frequency shift f_D , a timing offset $\Delta\tau$, and a frequency offset Δf between both stations due to manufacturing tolerances in the LOs. The FMCW transmit (tx) signals $s_{1k,tx}^{u/d}(t)$ and $s_{2k,tx}^{u/d}(t)$ are generated by direct digital synthesis (DDS1 and DDS2). The notation is chosen in the following way: The superscripts "u" and "d" denote up- and down-chirps. The subscripts "1" and "2" indicate signals at station 1 and 2, respectively. The subscript k defines the FMCW chirp number. More details will be given in Section III. The ADC in each station are named ADC1 and ADC2. Typically for FMCW radar, received (rx) signals $s_{1k,rx}^{u/d}(t)$ and $s_{2k,rx}^{u/d}(t)$ are down-converted with the locally generated signals by RF frequency mixers (M1 and M2). The mixing products are defined as $s_{1k,mix}^{u/d}(t)$ and $s_{2k,mix}^{u/d}(t)$ and the sampled signals are denoted by $s_{1k,beat}^{u/d}(t)$ and $s_{2k,beat}^{u/d}(t)$. The wireless channel between both stations is assumed to be reciprocal or quasi-reciprocal and time dependent, and the ToF is $\tau(t)$.

B. CFDDS-TWR Measurement Principle

A flowchart of all steps in the CFDDS-TWR algorithm is given in Fig. 2. The proposed technique requires the following operations.

1) *Presynchronization*: Prior to the radar measurement, wireless units must be presynchronized in time and frequency. After this step, the Nyquist sampling criterion is fulfilled for the beat signals in baseband and unambiguous reconstruction can be performed with the ADC signals $s_{1k,beat}^{u/d}(t)$ and $s_{2k,beat}^{u/d}(t)$. Presynchronization can be carried out by employing time triggered protocols or the method presented in [21]. By means of the latter technique, time and frequency offsets well below $|\Delta\tau| = 100$ ps and $|\Delta f| = 10$ Hz can be achieved. As the presented method does not require this degree of precision, we can simplify the presynchronization step by loosening the constraints to $|\Delta\tau| < 1$ μ s and

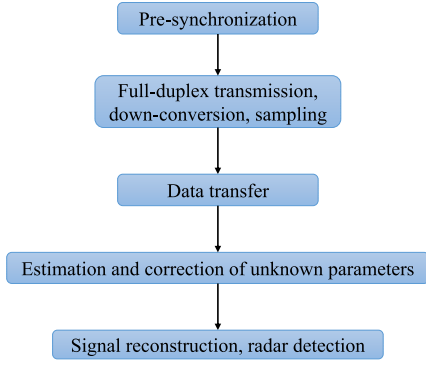


Fig. 2. Flowchart of the measurement sequence with CFDDS-TWR.

TABLE I
UPPER BOUNDS OF THE PRESYNCHRONIZATION PARAMETERS

Assumption	Parameter
$ \delta_t < 10^{-6}$	Relative drift in time
$ \Delta\tau < 1 \mu\text{s}$	Timing offset
$ \Delta f < 2 \text{ MHz}$	Frequency offset
$\tau(t) < 1 \mu\text{s}$	Time of flight

Assumptions on pre-synchronization quality for an FMCW secondary radar.

$|\Delta f| < 2 \text{ MHz}$. The relative LO drift is on the order of a few parts per million in practice [6]. We can therefore assume $|\delta_t| < 10^{-6}$ as the absolute value of the relative drift. Assuming a maximum range of 300 m, the ToF $\tau(t)$ does not exceed $1 \mu\text{s}$. And, the system works with a carrier frequency of $f_c = 24 \text{ GHz}$. A summary of the assumed upper bounds of the presynchronization parameters is given in Table I.

2) *FD Transmission, Down-Conversion, and Sampling*: The relative velocity cannot be measured by means of phase shift with a single up-chirp or an up-followed by a down-chirp, since frequency offset and Doppler shift are indistinguishable [6]. Doppler processing via secondary radar can be carried out by a sequence of interleaved up- and down-chirps. These types of chirp sequence FMCW secondary radar systems require additional information on the relative drift of the clock sources since the velocity to be measured is proportional to the phase shift in the “slow-time” domain. (Slow-time or interchirp time is defined in terms of the FMCW chirp number.) An FMCW chirp sequence is therefore transmitted by a full-duplex (FD) protocol and is simultaneously received and down-converted by both radar units.

3) *Data Transfer*: As the proposed algorithm requires ADC data from both radar stations, ADC data need to be transferred from at least one station. These data can be processed at one radar unit or by an external central processing unit (CPU).

4) *Estimating and Correcting Unknown Parameters*: The unknown system parameters δ_t , f_D , $\Delta\tau$, and Δf can be determined precisely from all the data. Corrections are carried out as these parameters impact the signal phases, the beat frequencies, and the PN level.

5) *Signal Reconstruction, Radar Detection*: Finally, this method outputs a synthesized beat signal, which is comparable

to a primary radar response. This signal is coherent and serves to suppress PN.

III. CFDDS-TWR SIGNAL MODEL WITH COARSE PRESYNCHRONIZATION

This mathematical model is given for line-of-sight (LOS) operation with two stations. The wireless radar units’ time bases are defined by their respective LOs. All operations such as the start of an FMCW chirp, frequency increment, and ADC timing rely on this time base. The S1 and S2 LO clock frequencies differ, as the stations are not physically linked.

Assuming local time increments generated by the LOs in stations 1 and 2 are $\Delta t_1 = 1/f_{\text{clk}1} = (1 + \delta_t/2)\Delta t$ and $\Delta t_2 = 1/f_{\text{clk}2} = (1 - \delta_t/2)\Delta t$, respectively,

$$t = \frac{t_1 + t_2}{2} = \frac{(1 + \delta_t/2)t + (1 - \delta_t/2)t}{2} \quad (1)$$

defines intrachirp time (often referred to as “fast time”) as the mean of the local time in each station. Parameter δ_t denotes the relative timing drift of station 1 with respect to station 2. Variables $f_{\text{clk}1}$ and $f_{\text{clk}2}$ denote the system reference clock at stations 1 and 2, respectively.

A. Mixed Signals With Full-Duplex Transmission

The phase of individual FMCW up- and down-chirps, generated at station 1, can be modeled as [5]

$$\begin{aligned} \Phi_{1k}^{u/d}(t_1) &= 2\pi \int_{-\infty}^{t_1} f_{c1}^{u/d} \pm \mu(t' + \Delta\tau_{1k})dt' + \Psi_{1k}(t_1) \\ &= 2\pi (f_{c1}^{u/d}t_1 \pm \mu(t_1^2/2 + \Delta\tau_{1k}t_1)) + \Psi_{1k}(t_1) + \Theta_{1k} \end{aligned} \quad (2)$$

where $f_{c1}^{u/d}$, μ , $\Delta\tau_{1k}$, $\Psi_{1k}(t_1)$, and Θ_{1k} denote the station 1 carrier frequency, chirp rate, timing offset, PN, and an arbitrary initial phase value, respectively. The upper superscript character “u” always represents up-chirps. The variable $k \in \{0, \dots, K-1\}$ indicates the FMCW chirp number, where K chirps are transmitted and received. Down-chirps are denoted by superscript “d.” The station 1 carrier frequency, $f_{c1}^{u/d} = f_c + \Delta f/2 \mp B/2$, consists of the common RF carrier frequency f_c , a frequency offset Δf in both stations that separates primary and secondary response, and the RF bandwidth B . Variables f_c , Δf , B , and μ are preset or default radar parameters, as in primary radar systems. The effect of $\Delta\tau_{1k}$ on the intrachirp PN is irrelevant since it can be modeled as stationary random process noise [22]. Replacing $f_{c1}^{u/d}$ in (2) yields

$$\begin{aligned} \Phi_{1k}^{u/d}(t) &= 2\pi (f_c + \Delta f/2 \mp B/2)(1 + \delta_t/2)t \\ &\quad \pm 2\pi \mu ((1 + \delta_t/2)^2/2t^2 + \Delta\tau_{1k}(1 + \delta_t/2)t) \\ &\quad + \Psi_{1k}((1 + \delta_t/2)t) + \Theta_{1k} \\ &= 2\pi (f_{c1}'^{u/d}t \pm \mu_1'/2t^2 \pm \mu \Delta\tau_{1k}'t) \\ &\quad + \Psi_{1k}((1 + \delta_t/2)t) + \Theta_{1k} \end{aligned} \quad (3)$$

where $f_{c1}'^{u/d} = (f_c + \Delta f/2 \mp B/2)(1 + \delta_t/2) = f_{c1}^{u/d}(1 + \delta_t/2)$, $\mu_1' = \mu(1 + \delta_t/2)^2 \approx \mu(1 + \delta_t)$, and $\Delta\tau_{1k}' = \Delta\tau_{1k}(1 + \delta_t/2)$ denote modified station 1 system parameters, which can be viewed as carrier frequency, chirp rate, and time delay for each individual chirp, dependent on the relative drift δ_t in timing.

The term $\mu\delta_t^2/4$ can be neglected for narrowband radar systems with slow chirp rates, assuming $|\delta_t| < 10^{-6}$.

Analogously, the up- and down-chirp phase at station 2 is derived as

$$\begin{aligned}\Phi_{2k}^{u/d}(t) &= 2\pi \int_{-\infty}^{t_2} f_{c2}^{u/d} \pm \mu(t' + \Delta\tau_{2k})dt' + \Psi_{2k}((1 - \delta_t/2)t) \\ &= 2\pi(f_c - \Delta f/2 \mp B/2)(1 - \delta_t/2)t \\ &\quad \pm 2\pi\mu((1 - \delta_t/2)^2/2t^2 + \Delta\tau_{2k}(1 - \delta_t/2)t) \\ &\quad + \Psi_{2k}((1 - \delta_t/2)t) + \Theta_{2k} \\ &= 2\pi(f_{c2}^{u/d'} \pm \mu'_2/2t^2 \pm \mu\Delta\tau'_{2k}t) \\ &\quad + \Psi_{2k}((1 - \delta_t/2)t) + \Theta_{2k}\end{aligned}\quad (4)$$

where $f_{c2}^{u/d} = f_c - \Delta f/2 \mp B/2$ is the RF carrier frequency for station 2 in terms of preset radar parameters. Accordingly, $f_{c2}^{u/d'} = (f_c - \Delta f/2 \mp B/2)(1 - \delta_t/2) = f_{c2}^{u/d}(1 - \delta_t/2)$, $\mu'_2 = \mu(1 - \delta_t/2)^2 \approx \mu(1 - \delta_t)$, and $\Delta\tau'_{2k} = \Delta\tau_{2k}(1 - \delta_t/2)$ denote the modified system parameters for station 2.

The instantaneous frequencies

$$\begin{aligned}f_{1k}^{u/d}(t) &= \frac{1}{2\pi} \frac{d}{dt} \Phi_{1k}^{u/d}(t) = f_{c1}^{u/d'} \pm \mu'_1 t \pm \mu\Delta\tau'_{1k} \\ &\quad + \frac{1}{2\pi} \frac{d}{dt} \Psi_{1k}((1 + \delta_t/2)t) \\ f_{2k}^{u/d}(t) &= f_{c2}^{u/d'} \pm \mu'_2 t \pm \mu\Delta\tau'_{2k} + \frac{1}{2\pi} \frac{d}{dt} \Psi_{2k}((1 - \delta_t/2)t)\end{aligned}\quad (5)$$

for stations 1 and 2 can be defined as functions of the auxiliary parameters in (3) and (4), respectively. The LO signals at stations 1 and 2

$$s_{1k,lo}^{u/d}(t) = s_{1k,tx}^{u/d}(t) = A_1^{u/d} \exp(j\Phi_{1k}^{u/d}(t))$$

and

$$s_{2k,lo}^{u/d}(t) = s_{2k,tx}^{u/d}(t) = A_2^{u/d} \exp(j\Phi_{2k}^{u/d}(t))\quad (6)$$

are transmitted as up- and down-chirps, and serve as references for down-conversion, where $A_1^{u/d}$ and $A_2^{u/d}$ denote the up- and down-chirp signal amplitudes, respectively.

Hence, the signals received at the opposite station over the time-dependent wireless channel with propagation time τ_k and attenuation coefficient α become

$$s_{2k,rx}^{u/d}(t) = \alpha A_1^{u/d} \exp(j\Phi_{1k}^{u/d}(t - \tau_k)) \exp(j2\pi f_D t)$$

and

$$s_{1k,rx}^{u/d}(t) = \alpha A_2^{u/d} \exp(j\Phi_{2k}^{u/d}(t - \tau_k)) \exp(j2\pi f_D t)\quad (7)$$

respectively, where the Doppler frequency f_D is assumed to be equal for stations 1 and 2. Since the FMCW chirp duration T_{sw} is relatively short (1 ms) and the ToF does not change rapidly, the propagation time τ_k in the wireless channel is assumed to depend only on the chirp number k . This condition ensures that the peak position does not change substantially during an FMCW chirp sequence. Otherwise, associating spectral peaks along a chirp sequence becomes challenging and direct range Doppler processing via 2-D FFT is impossible.

The mixed signals after down-conversion at station 1 can be modeled as

$$\begin{aligned}s_{1k,mix}^{u/d}(t) &= s_{1k,lo}^{u/d}(t) \cdot (s_{1k,rx}^{u/d}(t))^* \\ &= \alpha A_1^{u/d} A_2^{u/d} \exp\{j(2\pi(f_{c1}^{u/d'} t - f_{c2}^{u/d'}(t - \tau_k) - f_D t) \\ &\quad \pm 2\pi\mu(\Delta\tau'_{1k} t - \Delta\tau'_{2k}(t - \tau_k)) \\ &\quad \pm 2\pi(\mu'_1/2t^2 - \mu'_2/2(t - \tau_k)^2) \\ &\quad + \Psi_{1k}((1 + \delta_t/2)t) - \Psi_{2k}((1 - \delta_t/2)(t - \tau_k)) \\ &\quad + \Theta_{1k} - \Theta_{2k})\}\end{aligned}\quad (8)$$

after low-pass filtering in terms of the modified system parameters, where $()^*$ denotes complex conjugation. Alternatively, the mixed signals can be derived by substituting the exponential function with a cosine function. If in-phase and quadrature (I/Q) data are available, the complex conjugation in (8) must be applied to the first term. This signal is obtained by means of in-phase (I) sampling and subsequent Hilbert transform. Similarly

$$\begin{aligned}s_{2k,mix}^{u/d}(t) &= s_{2k,rx}^{u/d}(t) \cdot (s_{2k,lo}^{u/d}(t))^* \\ &= \alpha A_1^{u/d} A_2^{u/d} \exp\{j(2\pi(f_{c1}^{u/d'}(t - \tau_k) - f_{c2}^{u/d'} t + f_D t) \\ &\quad \pm 2\pi\mu(\Delta\tau'_{1k}(t - \tau_k) - \Delta\tau'_{2k} t) \\ &\quad \pm 2\pi(\mu'_1/2(t - \tau_k)^2 - \mu'_2/2t^2) \\ &\quad + \Psi_{1k}((1 + \delta_t/2)(t - \tau_k)) \\ &\quad - \Psi_{2k}((1 - \delta_t/2)t) + \Theta_{1k} - \Theta_{2k})\}\end{aligned}\quad (9)$$

expresses the IF signal at station 2. Without loss of generality, both IF frequencies in (8) and (9) are assumed positive, i.e., the modified carrier frequency $f_{c1}^{u/d'}$ is larger than $f_{c2}^{u/d'}$. This condition can be satisfied by choosing Δf positive and large enough. Please refer to [1] for a detailed discussion on the impact of frequency and timing offset on the beat mixed signals.

B. Chirp Sequence Modulation

We assume for the analysis of the chirp sequence modulation that stations 1 and 2 are delayed by an unknown timing offset $\Delta\tau$. Coarse presynchronization ensures that $\Delta\tau$ is not arbitrarily high. Therefore, FMCW chirps from both stations overlap most of the time and can be sampled according to the Nyquist criterion. In terms of the local time, the timing offsets across the slow time dimension

$$\begin{aligned}\Delta\tau_{1k} &= (1 + \delta_t/2)\Delta\tau/2 + \delta_t k T_{sw}/2 \approx \Delta\tau/2 + \delta_t k T_{sw}/2 \\ \Delta\tau_{2k} &= -(1 - \delta_t/2)\Delta\tau/2 - \delta_t k T_{sw}/2 \approx -\Delta\tau/2 - \delta_t k T_{sw}/2\end{aligned}\quad (10)$$

of the k th FMCW chirps of both stations are, respectively, a function of the relative timing drift δ_t , since the ramp trigger

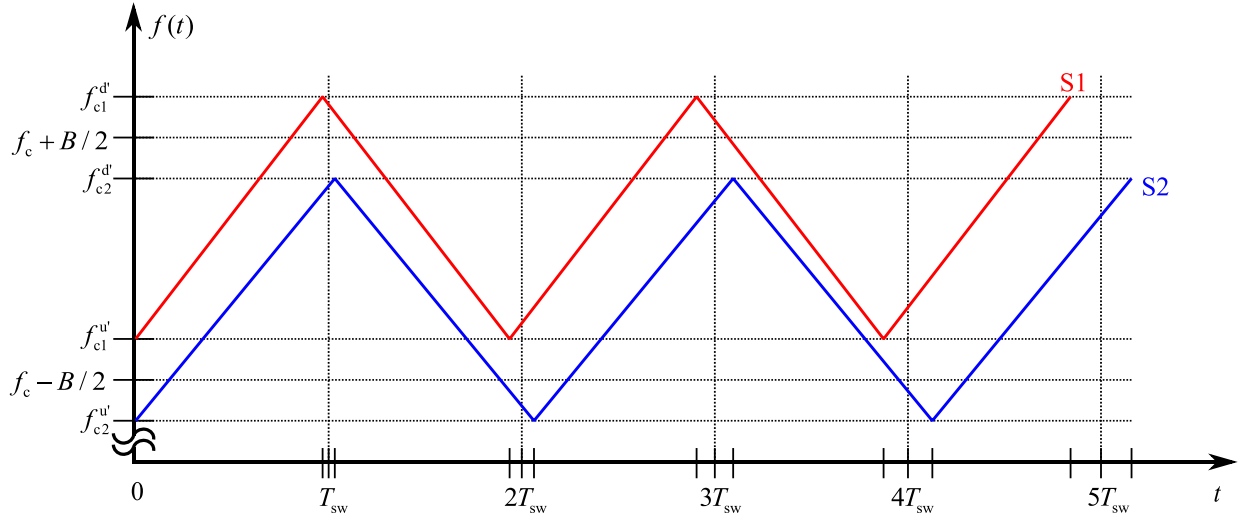


Fig. 3. Instantaneous frequencies of station 1 (S1) and station 2 (S2) as function of time with interleaved up- and down-chirps. For simplicity, the influence of random oscillator PN, Doppler frequency, timing offset, and frequency offset have been omitted. In practice, the influence of the drift in time is much smaller and the FMCW chirps of both stations overlap most of the time.

is controlled by the reference clock. Variable T_{sw} denotes the chirp duration. The local sweep time changes due to the timing drift. We simplify (10) by assuming that the product $\delta_t \Delta \tau \leq 1$ fs is negligible. This approximation makes both timing offsets equal in magnitude and of opposite sign such that

$$\Delta \tau_{1k} \approx \Delta \tau / 2 + \delta_t k T_{sw} / 2 \approx -\Delta \tau_{2k}. \quad (11)$$

Up- and down-chirps are interleaved such that they are indicated by even and odd k values, respectively. As $|\delta_t|$ was assumed to be on the order of 10^{-6} , $\Delta \tau'_{1k} = \Delta \tau_{1k}(1 + \delta_t/2) \approx \Delta \tau_{1k}$ and $\Delta \tau'_{2k} = \Delta \tau_{2k}(1 - \delta_t/2) \approx \Delta \tau_{2k}$ hold since both products $\delta_t \Delta \tau$ and $\delta_t^2 k T_{sw} / 4$ are negligible as they are smaller by many orders of magnitude than the other parts of (11).

The instantaneous frequencies of both stations in (5) are depicted in Fig. 3 for a sequence of interleaved up- and down-chirps for $K = 5$ chirps. PN and time and frequency offsets are omitted for better visualization. Both stations start transmission simultaneously at time $t = 0$. After the k th FMCW chirp, the chirp starts are delayed by $\Delta \tau_{1k} \approx -\delta_t k T_{sw} / 2$ and $\Delta \tau_{2k} \approx \delta_t k T_{sw} / 2$ in stations 1 and 2, respectively. In addition, the timing drift causes a slope different from the preset chirp rate μ across the fast time dimension.

C. Simplifying the Mixed Signals

For simplicity and clearness, we will analyze different terms in (8) and (9) separately. Employing basic calculus, we can simplify the first line in (8) to

$$\begin{aligned} \varphi_{a1k}^{u/d}(t) &= 2\pi \{f_{c1}^{u/d'} t - f_{c2}^{u/d'}(t - \tau_k) - f_D t\} \\ &= 2\pi \{(\Delta f + \delta_t(f_c \mp B/2) - f_D)t \\ &\quad + (f_c - \Delta f/2 \mp B/2)(1 - \delta_t/2)\tau_k\} \\ &\approx 2\pi \{(\delta_f - f_D)t + (f_c \mp B/2)\tau_k\} \end{aligned} \quad (12)$$

where the products $(f_c - \Delta f/2 \mp B/2)\delta_t \tau_k / 2$ and $\Delta f \tau_k / 2$ are orders of magnitude smaller than the other components

and can therefore be neglected. The total frequency offset is defined as $\delta_f = \Delta f + \delta_t(f_c \mp B/2)$. The phase $\varphi_{a1k}^{u/d}(t)$ depends on the difference of both modified carrier frequencies and the ToF τ_k in the wireless channel. Variable δ_f can be viewed as a time-independent frequency offset and $(f_c \mp B/2)\tau_k$ contains the phase information corresponding to the ToF. The second line in (8)

$$\begin{aligned} \varphi_{b1k}^{u/d}(t) &= \pm 2\pi \mu \{ \Delta \tau'_{1k} t - \Delta \tau'_{2k}(t - \tau_k) \} \\ &= \pm 2\pi \mu \{ (\Delta \tau + \delta_t k T_{sw})t \\ &\quad \times (-\Delta \tau + \delta_t(\Delta \tau + (\delta_t - 2)k T_{sw})/2)\tau_k / 2 \} \\ &\approx \pm 2\pi \mu (\Delta \tau + \delta_t k T_{sw})t \end{aligned} \quad (13)$$

depends on the chirp parameters and the timing drift. It can be simplified by neglecting the third line with the assumptions mentioned above. The phase $\varphi_{b1k}^{u/d}(t)$ is only comprised of the timing offset of each FMCW chirp, which is a function of the chirp number k . The third line in (8)

$$\begin{aligned} \varphi_{c1k}^{u/d}(t) &= 2\pi \{ \mu'_1 / 2 t^2 - \mu'_2 / 2 (t - \tau_k)^2 \} \\ &= \pm 2\pi \mu \{ (1 - \delta_t)\tau_k t + \delta_t t^2 - (1 - \delta_t)\tau_k^2 / 2 \} \\ &\approx \pm 2\pi \mu \{ \tau_k t + \delta_t t^2 \} \end{aligned} \quad (14)$$

is a function of the time of flight and the chirp rate difference due to the time drift. The first term of the last line in (14) contains the time of flight in the wireless channel, i.e., the classical information on range in FMCW radar. The second term in the last line in (14) is a quadratic function of time and is the same for all chirps. Simplifications are made based on the same assumptions as above.

According to (12)–(14), we simplify the first three lines of (9) to

$$\begin{aligned} \varphi_{a2k}^{u/d}(t) &\approx 2\pi \{ (\delta_f + f_D)t - (f_c \mp B/2)\tau_k \} \\ \varphi_{b2k}^{u/d}(t) &\approx \pm 2\pi \mu (\Delta \tau + \delta_t k T_{sw})t \end{aligned}$$

and

$$\varphi_{c2k}^{u/d}(t) \approx \pm 2\pi \mu \{ -\tau_k t + \delta_t t^2 \}. \quad (15)$$

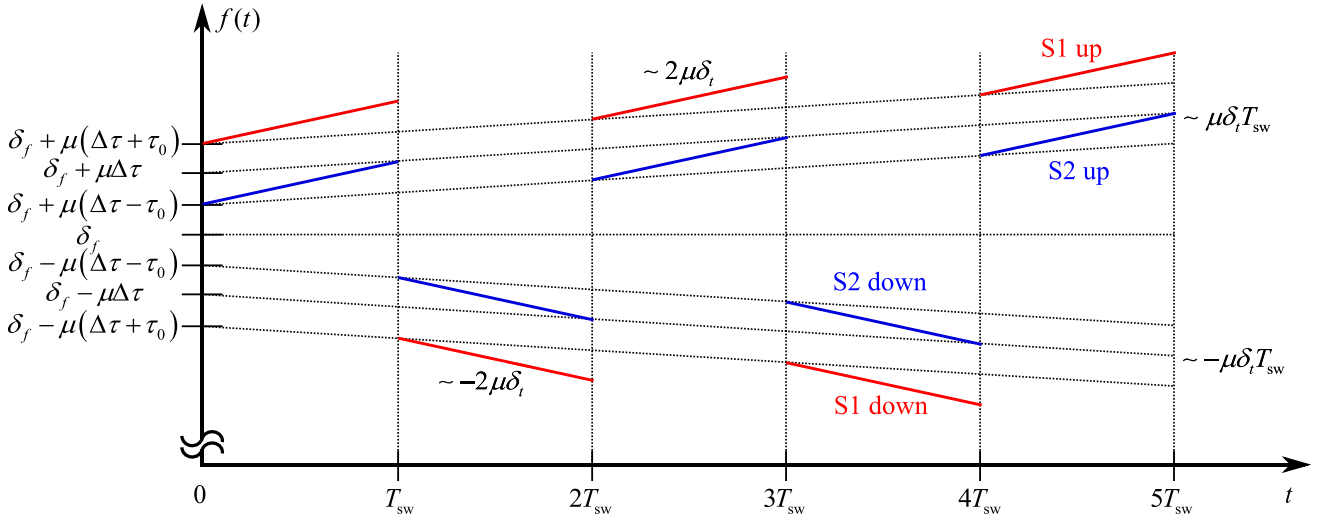


Fig. 4. Graphical representation of the instantaneous frequencies of the down-converted signals $s_{1k,\text{mix}}^{\text{u/d}}(t)$ and $s_{2k,\text{mix}}^{\text{u/d}}(t)$. PN and the Doppler frequency have been omitted for clarity.

With the results in (8) and (12)–(14), the mixed signal of station 1 can be stated as

$$\begin{aligned}
 s_{1k,\text{mix}}^{\text{u/d}}(t) &= \alpha A_1^{\text{u/d}} A_2^{\text{u/d}} \exp\{j(\varphi_{a1k}(t) + \varphi_{b1k}(t) + \varphi_{c1k}(t) \\
 &\quad + \Psi_{1k}((1 + \delta_t/2)t) - \Psi_{2k}((1 - \delta_t/2)(t - \tau_k)) \\
 &\quad + \Theta_{1k} - \Theta_{2k})\} \\
 &= A \exp\{j(2\pi((\delta_f - f_D)t + (f_c \mp B/2)\tau_k \\
 &\quad \pm \mu((\tau_0 + \Delta\tau + \delta_t k T_{\text{sw}})t + \delta_t t^2)) + \Theta_{1k} - \Theta_{2k} \\
 &\quad + \Psi_{1k}((1 + \delta_t/2)t) - \Psi_{2k}((1 - \delta_t/2)(t - \tau_k)))\} \quad (16)
 \end{aligned}$$

where $A = \alpha A_1^{\text{u/d}} A_2^{\text{u/d}}$ denotes the signal amplitude. Estimating the velocity by processing the phase shift requires the peak range position to remain almost constant. Hence, the ToF will be modeled as $\tau_k = \tau_0 + \tau'_k$, where τ_0 and τ'_k are the initial ToF and the propagation-time change in the wireless channel, respectively. Variable τ_k in the second line after the last equality sign becomes τ_0 . The station 2 mixed signal can be derived with the same assumptions as above as

$$\begin{aligned}
 s_{2k,\text{mix}}^{\text{u/d}}(t) &= \alpha A_1^{\text{u/d}} A_2^{\text{u/d}} \exp\{j(\varphi_{a2k}(t) + \varphi_{b2k}(t) + \varphi_{c2k}(t) \\
 &\quad + \Psi_{1k}((1 + \delta_t/2)(t - \tau_k)) \\
 &\quad - \Psi_{2k}((1 - \delta_t/2)t) + \Theta_{1k} - \Theta_{2k})\} \\
 &= A \exp\{j(2\pi((\delta_f + f_D)t - (f_c \mp B/2)\tau_k \\
 &\quad \pm \mu((-\tau_0 + \Delta\tau + \delta_t k T_{\text{sw}})t + \delta_t t^2)) + \Theta_{1k} - \Theta_{2k} \\
 &\quad + \Psi_{1k}((1 + \delta_t/2)(t - \tau_k)) - \Psi_{2k}((1 - \delta_t/2)t))\}. \quad (17)
 \end{aligned}$$

Comparing (16) and (17) reveals that all terms that contain the unknown system parameters δ_f , $\Delta\tau$, and δ_t have the same sign. The same holds for the unknown phase values caused by PN and initial LO phases. All terms which depend on τ_0 and τ_k have the opposite sign.

The instantaneous frequencies of the mixed signals in (16) and (17) with $f_D = 0$ are depicted in Fig. 4. The influence

of PN is neglected again as its impact on the instantaneous frequencies is negligible. After the k th FMCW chirp, the frequencies of the up- and down-chirps at the beginning of the fast time has increased by $2\mu\delta_t T_{\text{sw}}$ or reduced by $-2\mu\delta_t T_{\text{sw}}$. The up- and down-chirp frequencies increase and decrease by $2\mu\delta_t$ and $-2\mu\delta_t$, respectively, during each FMCW chirp.

D. Relationship Between Mixed and Sampled Signals

The ADCs in both radar units are controlled by their respective internal clock sources. Therefore, the mixing products in (16) and (17) are not available as sampled data. Sampling in stations 1 and 2 is delayed by $\Delta\tau_{1k}$ and $\Delta\tau_{2k}$, respectively. Besides this delay, the respective sampling intervals are not exactly equal due to the drift δ_t in timing. Since the drift is presumed to be relatively small, the sampling frequency is almost the same for both radar units. The above simplifications allow us to express the FMCW beat signals as a time delayed version of both mixed signals as

$$s_{1k,\text{beat}}^{\text{u/d}}(t) = s_{1k,\text{mix}}^{\text{u/d}}(t - \Delta\tau_{1k}) \approx s_{1k,\text{mix}}^{\text{u/d}}(t) \exp\{j\gamma_{1k}\}$$

and

$$s_{2k,\text{beat}}^{\text{u/d}}(t) = s_{2k,\text{mix}}^{\text{u/d}}(t - \Delta\tau_{2k}) \approx s_{2k,\text{mix}}^{\text{u/d}}(t) \exp\{j\gamma_{2k}\}. \quad (18)$$

This ADC delay causes the phase shift values

$$\gamma_{1k} \approx -2\pi\{(\delta_f - f_D)\Delta\tau_{1k} \pm \mu(\tau_0 + \Delta\tau + \delta_t k T_{\text{sw}})\Delta\tau_{1k}\}$$

and

$$\gamma_{2k} \approx -2\pi\{(\delta_f + f_D)\Delta\tau_{2k} \pm \mu(-\tau_0 + \Delta\tau + \delta_t k T_{\text{sw}})\Delta\tau_{2k}\} \quad (19)$$

which mainly depend on the mixed-signal frequencies from stations 1 and 2, respectively. Equation (19) is reduced to substantial terms, i.e., parts with a strong dependence on time.

These additional phase values can be avoided by physical triggering as in [18]. The timing offsets $\Delta\tau_{1k}$ and $\Delta\tau_{2k}$ are a function of the chirp number in wireless radar units and cause a time-dependent phase shift. Since there is no noticeable impact on the peak position, traditional FMCW secondary radar for

range estimation, which is based on frequency estimation, is not affected by this delay. However, the velocity information must be derived by evaluating the motion-induced phase change. Hence, the γ_{1k} and γ_{2k} values need to be corrected.

IV. CFDDS-TWR-BASED SYNCHRONIZATION AND SIGNAL RECONSTRUCTION

We shall now use the beat signals (18) from Section III to derive the estimation of the unknown system parameters δ_f , f_D , δ_t , and $\Delta\tau$ in this section. We shall then explain an algorithm for reconstructing a synthetic beat signal. Specifically, systematic errors are corrected and PN reduced by fusing data from both radar units after digital postprocessing.

A. Determining Unknown Parameters

Theoretically, the offsets $\Delta\tau_{1k}$ and $\Delta\tau_{2k}$ in time can be determined for each individual FMCW chirp by comparing up- and down-chirps, as stated in [6]. Due to chirp sequence modulation, the timing drift δ_t generates another unknown component and a variation of the beat signals in the slow time domain. Furthermore, the signal-to-noise ratio (SNR) is much lower when conducting an estimation with individual combinations of FMCW chirps and the quadratic expression $\delta_t t^2$ causes a broadening of the spectrum and complicates peak search.

To overcome these drawbacks, we shall employ a method closely related to spectral line search with beat signals in the frequency domain. As both signals in (18) are only available as sampled signals, we must use a discrete Fourier transform. Since unknown parameters can be derived via discretized or continuous signals if the Nyquist sampling criterion holds, all computational steps are described with continuous time signals. FMCW chirps are of limited duration $t < T_{sw}$ in fast time. Therefore, a suitable window function $w(t)$ is multiplied with the time domain signals to reduce the sidelobe level in the frequency domain. First, the Fourier transform $\mathcal{F}\{\cdot\}$ of the first line in (18) after windowing is calculated as

$$\begin{aligned} S_{1k,beat}^{u/d}(f) &= \mathcal{F}\{w(t)s_{1k,beat}^{u/d}(t)\} \\ &= A\delta\{f - ((\delta_f - f_D) \pm \mu(\tau_0 + \Delta\tau + \delta_t k T_{sw}))\} \\ &\quad * \mathcal{F}\{w(t) \exp\{\pm j2\pi\mu\delta_t t^2\}\} \\ &\quad * \mathcal{F}\{\exp\{j\Psi_{pn,1k}(t)\} \cdot \exp\{j\gamma_{1k}\} \\ &\quad \cdot \exp\{j(2\pi(f_c \mp B/2)\tau_k + \Theta_{1k} - \Theta_{2k})\} \end{aligned} \quad (20)$$

where $\delta\{\cdot\}$ and $*$ denote the Dirac delta distribution and the convolution operator, respectively. We abbreviate the expression for the PN of the k th FMCW chirp to $\Psi_{pn,1k}(t) = \Psi_{1k}((1 + \delta_t/2)t) - \Psi_{2k}((1 - \delta_t/2)(t - \tau_k))$; it has negligible influence on the peak position [23]. Hence, it can be approximated by $\mathcal{F}\{\exp\{j\Psi_{pn,1k}(t)\}\} \approx \delta\{f\}$ to estimate the unknown parameters. The third line in (20) is the Fourier transform of the window function multiplied by a nonlinear chirp, as described in [24]. This function causes substantial broadening of spectral peaks.

Using the abbreviation $\Psi_{pn,2k}(t) = \Psi_{1k}((1 + \delta_t/2)(t - \tau_k)) - \Psi_{2k}((1 - \delta_t/2)t)$ and the abovementioned approximation $\mathcal{F}\{\exp\{j\Psi_{pn,2k}(t)\}\} \approx \delta\{f\}$, we can express the beat

spectrum for the second radar station as

$$\begin{aligned} S_{2k,beat}^{u/d}(f) &= A\delta\{f - ((\delta_f + f_D) \pm \mu(-\tau_0 + \Delta\tau + \delta_t k T_{sw}))\} \\ &\quad * \mathcal{F}\{w(t) \exp\{\pm j2\pi\mu\delta_t t^2\}\} \\ &\quad * \mathcal{F}\{\exp\{j\Psi_{pn,2k}(t)\} \cdot \exp\{j\gamma_{2k}\} \\ &\quad \cdot \exp\{j(-2\pi(f_c \mp B/2)\tau_k + \Theta_{1k} - \Theta_{2k})\}. \end{aligned} \quad (21)$$

As a first step in estimating the unknown parameters δ_f , f_D , δ_t , and $\Delta\tau$, we need to, respectively, analyze the absolute values

$$\begin{aligned} \hat{S}_{1k,beat}^{u/d}(f) &= |S_{1k,beat}^{u/d}(f)| \\ &= A\delta\{f - ((\delta_f - f_D) \pm \mu(\tau_0 + \Delta\tau + \delta_t k T_{sw}))\} \\ &\quad * |\mathcal{F}\{w(t) \exp\{\pm j2\pi\mu\delta_t t^2\}\}| \end{aligned} \quad (22)$$

and

$$\begin{aligned} \hat{S}_{2k,beat}^{u/d}(f) &= |S_{2k,beat}^{u/d}(f)| \\ &= A\delta\{f - ((\delta_f + f_D) \pm \mu(-\tau_0 + \Delta\tau + \delta_t k T_{sw}))\} \\ &\quad * |\mathcal{F}\{w(t) \exp\{\pm j2\pi\mu\delta_t t^2\}\}| \end{aligned} \quad (23)$$

of the beat spectra of units 1 and 2 in (20) and (21). Both spectra contain the spectral mask $\mathcal{F}\{w(t) \exp\{\pm j2\pi\mu\delta_t t^2\}\}$, which does not depend on the chirp number k . The location of this spectral mask is a linear function of the timing drift δ_t . We can therefore determine the drift estimate $\hat{\delta}_t$ by tracking the location of the unknown spectral mask, i.e., by means of a line search in a 2-D image. Numerous well-known methods are described in [25].

Alternatively, the magnitude spectra of (22) and (23) can be divided into two halves. We estimate the timing drifts with

$$\begin{aligned} \hat{\delta}_t &= \pm \frac{2}{KT_{sw}} \\ &\quad \times \arg \max_f \left\{ \sum_{k=1}^{K/2} \text{corr}\{\hat{S}_{1k,beat}^{u/d}(f), \hat{S}_{1(k+K/2),beat}^{u/d}(f)\} \right\} \end{aligned} \quad (24)$$

where $\text{corr}\{X(f), Y(f)\} = \int_{-\infty}^{\infty} X(f')Y(f' + f)df'$ defines the cross correlation of continuous functions. Again, the upper superscript symbol signifies up-chirps. Symbols with hats are estimated values. Equation (24) can be evaluated with both beat spectra and with up- and down-chirps and averaged to increase the accuracy. Since δ_t defines the quadratic function in (16) and (17) as well as the peak position of every chirp, both effects must be compensated for by multiplication in the time domain and by frequency shifting, respectively. After these corrections

$$\hat{S}_{1k,beat}^{u/d}(f) \approx A\delta\{f - ((\delta_f - f_D) \pm \mu(\tau_0 + \Delta\tau))\} * W(f)$$

and

$$\hat{S}_{2k,beat}^{u/d}(f) \approx A\delta\{f - ((\delta_f + f_D) \pm \mu(-\tau_0 + \Delta\tau))\} * W(f) \quad (25)$$

define the spectra of the first and second beat signals, respectively. Both spectra are mainly shaped by the Fourier transform of the window function. Again, the effect of PN on the

spectral mask $W(f)$ is neglected. Both functions in (25) are independent of k and the beat frequencies can be obtained by peak search after averaging over all k , regardless of the additional timing offset caused by interleaving up- and down-chirps, as shown in (25). Thereafter, the beat frequencies

$$\begin{aligned} f_{1,\text{beat}}^{u/d} &= \arg \max_f \frac{1}{K} \sum_k \hat{S}_{1k,\text{beat}}^{u/d}(f) = \delta_f - f_D \pm \mu(\tau_0 + \Delta\tau) \\ f_{2,\text{beat}}^{u/d} &= \arg \max_f \frac{1}{K} \sum_k \hat{S}_{2k,\text{beat}}^{u/d}(f) = \delta_f + f_D \pm \mu(-\tau_0 + \Delta\tau) \end{aligned} \quad (26)$$

of the up- and down-chirps of both beat spectra yield a system of linear equations comparable to the result for the synchronization parameters in [6]. The frequency offset is given by

$$\hat{\delta}_f = (f_{1,\text{beat}}^u + f_{1,\text{beat}}^d + f_{2,\text{beat}}^u + f_{2,\text{beat}}^d)/4 \quad (27)$$

the initial timing offset by

$$\Delta\hat{\tau} = (f_{1,\text{beat}}^u - f_{1,\text{beat}}^d + f_{2,\text{beat}}^u - f_{2,\text{beat}}^d)/4 \quad (28)$$

and the Doppler frequency by

$$\hat{f}_D = (-f_{1,\text{beat}}^u - f_{1,\text{beat}}^d + f_{2,\text{beat}}^u + f_{2,\text{beat}}^d)/4. \quad (29)$$

As stated in [17], the ToF can be estimated by comparing the beat frequencies of the up-chirps of both beat spectra in (25), if δ_t is negligible.

B. Reconstruction Algorithm

Having estimated all unknown parameters, we derive a single synthetic radar signal. Equations (20) and (21) show the respective complex beat spectra of both wireless units. We will explain below the correction of all terms that are independent of the ToF $\tau_k = \tau_0 + \tau'_k$.

We need to compensate the ADC delay in (18) to eliminate the unknown phase shift of every chirp using the systematic effects defined in (24) and (28) to achieve coherent operation with free running oscillators. To this end, we estimate the individual timing offset of every FMCW chirp at stations 1 and 2 as

$$\Delta\hat{\tau}_{1k} = -\Delta\hat{\tau}_{2k} = \Delta\hat{\tau}/2 + \hat{\delta}_t k T_{\text{sw}}/2. \quad (30)$$

Now, we correct the respective beat signal delays in (18) by applying time shifting, i.e., by resampling the discrete data. This can be done by shifting sampled data by an integer value and subsequently applying a fractional delay filter. Since the phase shift values γ_{1k} and γ_{2k} depend on the frequency of the mixed signals, correction in the time domain always works.

After the compensation, all terms that are a function of $\tau_k = \tau_0 + \tau'_k$ are complex conjugated as in (16) and (17). The quadratic component must be removed in the next step by multiplying the beat signal in the time domain by a function of the estimate $\hat{\delta}_t$ of the timing drift. The Fourier transform

after this operation yields

$$\begin{aligned} \tilde{S}_{1k,\text{mix}}^{u/d}(f) &= \mathcal{F}\{s_{1k,\text{mix}}^{u/d}(t) \times \exp\{\mp j2\pi\mu\hat{\delta}_t t^2\}\} \\ &\approx A\delta\{f - ((\delta_f - f_D) \pm \mu(\tau_0 + \Delta\tau + \hat{\delta}_t k T_{\text{sw}}))\} \\ &\quad * \mathcal{F}\{\exp\{j\Psi_{\text{pn},1k}(t)\}\} * W(f) \\ &\quad \times \exp\{j(2\pi(f_c \mp B/2)\tau_k + \Theta_{1k} - \Theta_{2k})\} \end{aligned} \quad (31)$$

with the same notation as in (20). Similarly

$$\begin{aligned} \tilde{S}_{2k,\text{mix}}^{u/d}(f) &\approx A\delta\{f - ((\delta_f + f_D) \pm \mu(-\tau_0 + \Delta\tau + \hat{\delta}_t k T_{\text{sw}}))\} \\ &\quad * \mathcal{F}\{\exp\{j\Psi_{\text{pn},2k}(t)\}\} * W(f) \\ &\quad \times \exp\{j(-2\pi(f_c \mp B/2)\tau_k + \Theta_{1k} - \Theta_{2k})\} \end{aligned} \quad (32)$$

describes the corrected beat signal of the second mixing product.

All remaining steps will be performed with (31) and (32). For each FMCW chirp, the beat spectra must be compensated for with the frequency shift

$$f_{\text{shift},1k}^{u/d} = \hat{\delta}_f - f_D \pm \mu(\Delta\hat{\tau} + \hat{\delta}_t k T_{\text{sw}})$$

and

$$f_{\text{shift},2k}^{u/d} = \hat{\delta}_f + f_D \pm \mu(\Delta\hat{\tau} + \hat{\delta}_t k T_{\text{sw}}). \quad (33)$$

This step yields

$$\begin{aligned} \tilde{\tilde{S}}_{1k,\text{mix}}^{u/d}(f) &\approx A\delta\{f \mp \mu\tau_0\} * \mathcal{F}\{\exp\{j\Psi_{\text{pn},1k}(t)\}\} * W(f) \\ &\quad \times \exp\{j(2\pi(f_c \mp B/2)\tau_k + \Theta_{1k} - \Theta_{2k})\} \end{aligned}$$

and

$$\begin{aligned} \tilde{\tilde{S}}_{2k,\text{mix}}^{u/d}(f) &\approx A\delta\{f \pm \mu\tau_0\} * \mathcal{F}\{\exp\{j\Psi_{\text{pn},2k}(t)\}\} * W(f) \\ &\quad \times \exp\{j(-2\pi(f_c \mp B/2)\tau_k + \Theta_{1k} - \Theta_{2k})\} \end{aligned} \quad (34)$$

for the beat spectra of stations 1 and 2, respectively. We find frequency-domain peaks now at $\mp\mu\tau_0$, which corresponds to the channel length. The corresponding signals in the time domain are $\tilde{\tilde{s}}_{1k,\text{mix}}^{u/d}(t)$ and $\tilde{\tilde{s}}_{2k,\text{mix}}^{u/d}(t)$ for stations 1 and 2, respectively.

Unknown parameters impacting the beat frequency in (34) have effectively been corrected at this point. We need to analyze the PN of both signals in this equation to accurately correct the unknown parameters. Based on the above definition, the expressions

$$\begin{aligned} \Psi_{\text{pn},1k}(t) &= \Psi_{1k}((1 + \delta_t/2)t) - \Psi_{2k}((1 - \delta_t/2)(t - \tau_k)) \\ \Psi_{\text{pn},2k}(t) &= \Psi_{1k}((1 + \delta_t/2)(t - \tau_k)) - \Psi_{2k}((1 - \delta_t/2)t) \end{aligned} \quad (35)$$

give the PN—a stochastic process. These stochastic formulas are not exactly equal; they are delayed by the ToF τ_k . Generally, this influence causes a broadening and impairs the spectral-peak search. The respective phase differences of both values in (35) correspond to the magnitude of the oscillator PN profile at the position $f_{\text{pn}} \sim 1/\tau_k$. These profiles typically feature a decay with increasing frequency and an influence >1 MHz, which corresponds to the assumed maximum of 1 μs in ToF, which is assumed to be negligible.

As shown in [26], these random variables can be approximated via Taylor expansion by

$$\exp\{j\Psi_{\text{pn},1k}(t)\} \approx \exp\{j\Psi_{\text{pn},2k}(t)\} \approx 1 + j\varepsilon_k(t) \quad (36)$$

where the residual PN $\varepsilon_k(t) \approx \Psi_{\text{pn},1k}(t) \approx \Psi_{\text{pn},2k}(t)$ in (35) is comparable in both mixing products.

The first approximation in (36) is valid since the PN causes almost the same phase shift in both parts of (34) due to the short ToF. The second equation approximates the PN as a function of the linear part $\varepsilon_k(t)$ of this Taylor expansion. As the initial phase of every chirp is modeled as Θ_{1k} and Θ_{2k} , (36) is a stochastic process with zero mean.

With this knowledge, estimates of the phase shift can be obtained as phases

$$\phi_k = \left(\arg \left\{ \tilde{s}_{1k,\text{mix}}^{\text{u/d}} \left(\arg \max_f \{ |\tilde{s}_{1k,\text{mix}}^{\text{u/d}}(f)| \} \right) \right\} \right. \\ \left. + \arg \left\{ \tilde{s}_{2k,\text{mix}}^{\text{u/d}} \left(\arg \max_f \{ |\tilde{s}_{2k,\text{mix}}^{\text{u/d}}(f)| \} \right) \right\} \right) / 2 \quad (37)$$

at the peak value of (34), where $\arg\{\cdot\}$ denotes the argument of a complex scalar. Having subtracted phase ϕ_k from (34) by multiplying it by the function $\exp\{-j\phi_k\}$, the phases are individually unwrapped along the chirp number k for up- and down-chirps due to the division by factor 2 in (37). To account for this ambiguity, the residual phase of the first up- and down-chirp is denoted by $\phi_0^{\text{u/d}} \in m\pi$ with $m \in \mathbb{Z}$. This operation corrects the unknown initial phase values Θ_{1k} and Θ_{2k} for all FMCW chirps but the first. Performing a complex conjugation of $\tilde{s}_{1k,\text{mix}}^{\text{u/d}}(t)$ and $\tilde{s}_{2k,\text{mix}}^{\text{u/d}}(t)$ after subtracting ϕ_k ensures that the ToF causes the same phase shift in all spectra. The spectral peaks of the up-chirps in station 1 and the down-chirps in station 2 can now be found at $\mu\tau_0$. After these processing steps, both spectra can be represented as

$$\tilde{s}_{1k,\text{mix}}^{\text{u/d}}(f) \approx A\delta\{f - \mu\tau_0\} * \mathcal{F}\{1 \pm j\varepsilon_k(t)\} * W(f) \\ \times \exp\{\pm j2\pi(f_c \mp B/2)\tau_k\} \exp\{\pm j\phi_0^{\text{u/d}}\}$$

and

$$\tilde{s}_{2k,\text{mix}}^{\text{u/d}}(f) \approx A\delta\{f - \mu\tau_0\} * \mathcal{F}\{1 \mp j\varepsilon_k(t)\} * W(f) \\ \times \exp\{\pm j2\pi(f_c \mp B/2)\tau_k\} \exp\{\mp j\phi_0^{\text{u/d}}\}. \quad (38)$$

We retrieve the initial phase by unwrapping

$$\hat{\phi}_0^{\text{u/d}} = U \left\{ \arg \left\{ \tilde{s}_{10,\text{mix}}^{\text{u/d}} \left(\arg \max_f \{ |\tilde{s}_{10,\text{mix}}^{\text{u/d}}(f)| \} \right) \right\} \right. \\ \left. - \arg \left\{ \tilde{s}_{20,\text{mix}}^{\text{u/d}} \left(\arg \max_f \{ |\tilde{s}_{20,\text{mix}}^{\text{u/d}}(f)| \} \right) \right\} \right\} \\ \approx 4\pi(f_c \mp B/2)\tau_0 \bmod 2\pi \quad (39)$$

where $U\{\cdot\}$ and $\bmod$ denote the phase unwrapping and modulo operations, respectively. After applying (39), we obtain the initial phase in an unambiguous range comparable to a primary radar system, i.e., it scales with the round-trip ToF instead of the one-way ToF.

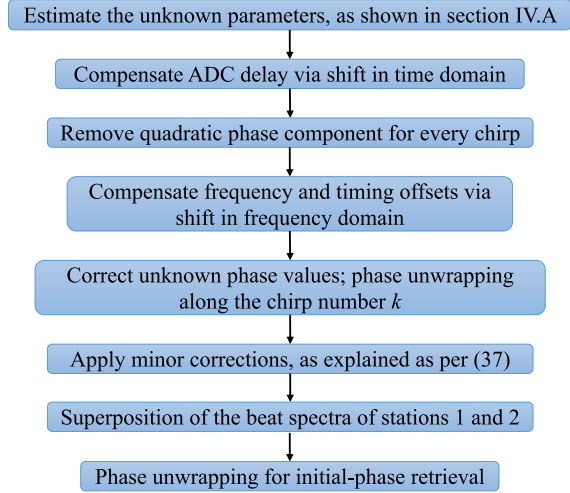


Fig. 5. Summary of the reconstruction algorithm.

C. Phase Noise Suppression via Superposition

In the final step of the algorithm, the superposition

$$S_{k,\text{synth}}^{\text{u/d}}(f) = (\tilde{s}_{1k,\text{mix}}^{\text{u/d}}(f) + \tilde{s}_{2k,\text{mix}}^{\text{u/d}}(f)) / 2 \\ = A\delta\{f - \mu\tau_0\} * W(f) \\ \times \exp\{\pm j2\pi(f_c \mp B/2)\tau_k\} \exp\{j\phi_0^{\text{u/d}}\} \quad (40)$$

delivers the beat spectra, which are compensated for intrachirp PN. In practice, the amplitudes of the mixed signals in both stations are not necessarily equal and must be normalized prior to superposition. As the Fourier transform is a linear operation, the effect of $\mathcal{F}\{j\varepsilon_k(t)\}$ is canceled out by this superposition due to opposite signs in the two beat spectra. The phases of the sequences of up- and down-chirps can be evaluated individually. A summary of the reconstruction algorithm is given in Fig. 5.

The largest gain of the PN suppression occurs when the PN level is much higher than the thermal noise floor. Due to a power decay of $1/\tau^2$ and individual LOs, this condition is satisfied for indoor applications at ranges below 100 m. As the superposition is based on the linear approximation in (36), the remaining error in

$$\exp\{j\varepsilon_k(t)\} \approx 1 + j\varepsilon_k(t) - O(\varepsilon_k^2(t)) \quad (41)$$

is in the order of the quadratic residual $\varepsilon_k^2(t)$, where $O(\cdot)$ expresses the O-notation. Hence, the quality of PN suppression greatly depends on the PN profile of the LOs.

D. Range and Velocity Estimation and Phase Tracking

Up to this point, the ToF in the wireless channel was modeled as $\tau_k = \tau_0 + \tau'_k$, where τ_0 is the initial ToF and τ'_k is the change in ToF, which is assumed to be relatively small. Since knowledge of range and velocity instead of ToF are required in most use cases, we express linear motion during an FMCW burst as

$$x_k = c_0\tau_k = x_0 + kvT_{\text{sw}} \quad (42)$$

where c_0 , x_0 , and v denote the phase velocity in free space, the initial position, and the radial velocity, respectively.

With (42), we express the 2-D Fourier transform of the synthetic beat signal as

$$S_{k,\text{synth}}^{u/d}(f, f_2) \approx A\delta\{f - \mu x_0/c_0\} * W(f) \cdot \delta\{f_2 \pm (f_c \mp B/2)vT_{\text{sw}}/c_0\} * W_2(f_2) \times \exp\{\pm j2\pi(f_c \mp B/2)x_0/c_0\} \exp\{j\phi_0^{u/d}\} \quad (43)$$

where f_2 and W_2 denote the frequency along the slow time dimension and the corresponding window function in this dimension, respectively.

The resulting unambiguous range, by contrast to primary FMCW radar systems, is not only a function of sweep rate and sampling frequency. Time and frequency offsets offer two degrees of freedom to determine the maximum range of these systems. Details on the implementation and calculations are given in [27].

For this method, we calculate the unambiguous velocity for up- and down-chirps as

$$v_{\text{max}}^{u/d} = c_0/(4(f_c \mp B/2)T_{\text{sw}}). \quad (44)$$

It can be derived by restricting the allowed phase shift along slow time dimension in (40) to π and inserting τ_k from (42). This is one half (compared to a primary radar system), as interleaving reduces the sampling rate along the slow time dimension. In addition, the phase-unwrapping requirement reduces the detectable phase shift by a factor of two.

Phase tracking is known from applications with primary continuous wave (CW) radar systems [28], [29]. A small spatial displacement Δx is estimated by evaluating the initial phase of every FMCW burst in (39). Using this equation, we express the phase difference as

$$\Delta\hat{\phi}_0^{u/d} = 4\pi(f_c \mp B/2)\Delta x/c_0 \quad (45)$$

where phase unwrapping guarantees unambiguous recovery of this difference. As the unambiguous range scales with the RF carrier frequency, only round-trip displacements smaller than a quarter carrier wavelength can be detected, as $\Delta\hat{\phi}_0^{u/d}$ must not exceed $\pm\pi/2$. The phase shift in narrowband radar systems can be estimated with greater sensitivity than the difference in beat frequency of consecutive FMCW bursts.

V. CFDDS-TWR WITH PRECISE PRESYNCHRONIZATION

Since the algorithm summarized in Table II is quite complex, a simplified signal model with accurate presynchronization of the drift in timing will be explained in this section. Range and phase estimation with two radar units that underlay a strong drift in timing require a large IF bandwidth, see Fig. 4, since the instantaneous frequencies of the down-converted signals depend on the chirp number k . As all unknown system parameters can be determined very accurately by the method described in Section IV-A, the LO mismatch of either one or both radar units can be compensated for this drift to ensure that the LO frequencies f_{clk1} and f_{clk2} are the same. Furthermore, the unambiguous Doppler is doubled by employing up-chirps instead of up- and down-chirps as the interchirp time can theoretically be one half.

TABLE II
SIMULATION AND TEST PARAMETERS

Symbol	Parameter	Value
f_c	RF carrier frequency	24 GHz
B	RF bandwidth	250 MHz and 500 MHz
T_{sw}	Chirp duration	1 ms
μ	Chirp slope	250 GHz/s and 500 GHz/s
K	Number of chirps	64
N_r	Number of range bins	2^{14}
N_v	Number of velocity bins	2^9

Radar Parameters for simulations and tests.

A. Signal Model

The clock frequency of a voltage-controlled crystal oscillator (VCXO) can be tuned very accurately in practice [30]. After the estimation of the relative drift in time $\hat{\delta}_t$ with the method in IV.A, the LO frequency $f_{\text{clk1/2}}$ of at least one station can be adjusted. We can therefore ensure that $\delta_t \approx 0$ and that the timing offset $\Delta\tau_{1k} \approx \Delta\tau/2 \approx -\Delta\tau_{2k}$ in (10) is the same for every FMCW chirp. The Doppler frequency is assumed to be very low, i.e., $f_D \approx 0$ Hz. Alternatively, the influence of f_D on the ranging accuracy can be corrected after the velocity estimation. Assuming $\delta_t = 0$ and $\delta_f = \Delta f$, we can simplify the beat spectra in (20) and (21) to

$$S_{1k,\text{beat}}^{\text{sync}}(f) = A\delta\{f - (\Delta f + \mu(\tau_0 + \Delta\tau))\} * W(f) * \mathcal{F}\{\exp\{j\Psi_{\text{pn},1k}(t)\}\} \cdot \exp\{j\gamma_1\} \cdot \exp\{j(2\pi(f_c - B/2)\tau_k + \Theta_{1k} - \Theta_{2k})\} \quad (46)$$

for station 1 and

$$S_{2k,\text{beat}}^{\text{sync}}(f) = A\delta\{f - (\Delta f + \mu(-\tau_0 + \Delta\tau))\} * W(f) * \mathcal{F}\{\exp\{j\Psi_{\text{pn},2k}(t)\}\} \cdot \exp\{j\gamma_2\} \cdot \exp\{j(-2\pi(f_c - B/2)\tau_k + \Theta_{1k} - \Theta_{2k})\} \quad (47)$$

for station 2, where the superscript “sync” signifies precise presynchronization. The corresponding signals in the time domain are defined as $s_{1k,\text{beat}}^{\text{sync}}(t)$ and $s_{2k,\text{beat}}^{\text{sync}}(t)$. Without the drift in timing and slow movement, the spectral peak positions do not vary along k . All phase values of opposite sign in (46) and (47) are a function only of the ToF. These spectral peaks are centered around the common frequency shift $f_{\text{shift}} = \Delta f + \mu\Delta\tau$, which is comprised of the preset Δf and the time delay $\Delta\tau$ for stations 1 and 2. Again, the sampling criterion must be satisfied for the beat frequency maximum at $\Delta f + \mu(\tau_0 + \Delta\tau)$. In contrast to the beat spectra in (20) and (21), there is no quadratic component and spectral broadening anymore and the peak search can run directly.

The instantaneous frequencies of the beat signals are depicted in Fig. 6. The beat frequencies of the down-converted signals in stations 1 and 2 have the common center f_{shift} for all individual FMCW chirps. The difference of both frequencies is proportional to the round-trip ToF.

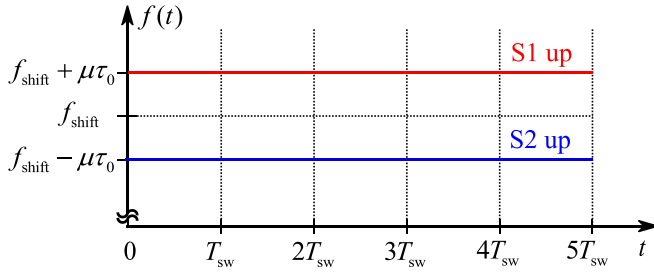


Fig. 6. Instantaneous frequencies of the beat signals $s_{1k,beat}^{sync}(t)$ and $s_{2k,beat}^{sync}(t)$ with precise synchronization of the offset in timing. The influence of PN has been omitted.

B. LOS Estimation of ToF and Phase

These equations are like the model in [17] for a single FMCW chirp. Therefore, we estimate the ToF directly and individually for each FMCW chirp by calculating

$$\hat{\tau}_{0,k} = \left(\arg \max_f \{S_{1k,beat}^{sync}(f)\} - \arg \max_f \{S_{2k,beat}^{sync}(f)\} \right) / (2\mu). \quad (48)$$

PN is thus suppressed. Different from [17], the estimation accuracy is enhanced greatly by averaging

$$\hat{\tau}_0 = \frac{1}{K} \sum_k \hat{\tau}_{0,k} \quad (49)$$

along the chirp number k . We estimate the phase change along the chirp sequence directly as well by evaluating

$$\begin{aligned} \hat{\phi}_k &= \arg \{S_{1k,beat}^{sync}(\arg \max_f \{|S_{1k,beat}^{sync}(f)|\})\} \\ &\quad - \arg \{S_{2k,beat}^{sync}(\arg \max_f \{|S_{2k,beat}^{sync}(f)|\})\} \\ &= 4\pi(f_c - B/2)\tau_k + \phi_0 \end{aligned} \quad (50)$$

where $\phi_0 = \gamma_1 - \gamma_2$ is a function of the unknown ADC delay caused by the timing offset. We then convert ToF and phase variation to range and velocity, as shown in Section IV-D.

C. Superposition

Another option is to calculate a single synthetic beat spectrum, like (40). Here, we estimate the frequency shift by computing

$$\begin{aligned} \hat{f}_{shift} &= \frac{1}{K} \sum_k \left(\arg \max_f \{|S_{1k,beat}^{sync}(f)|\} \right. \\ &\quad \left. + \arg \max_f \{|S_{2k,beat}^{sync}(f)|\} \right) \\ &\approx \Delta f + \mu \Delta \tau. \end{aligned} \quad (51)$$

This shift is constant along each FMCW chirp k and can be corrected for every chirp. We compensate $\hat{f}_{shift}$ by spectral shifting and express the resulting beat spectra as

$$\begin{aligned} \tilde{S}_{1k,beat}^{sync}(f) &\approx A\delta(f - \mu\tau_0) * W(f) \\ &\quad * \mathcal{F}\{\exp\{j\Psi_{pn,1k}(t)\} \times \exp\{j\gamma_1\} \\ &\quad \times \exp\{j(2\pi(f_c - B/2)\tau_k + \Theta_{1k} - \Theta_{2k})\} \end{aligned} \quad (52)$$

for station 1 and

$$\begin{aligned} \tilde{S}_{2k,beat}^{sync}(f) &\approx A\delta(f + \mu\tau_0) * W(f) \\ &\quad * \mathcal{F}\{\exp\{j\Psi_{pn,2k}(t)\} \times \exp\{j\gamma_2\} \\ &\quad \times \exp\{j(-2\pi(f_c - B/2)\tau_k + \Theta_{1k} - \Theta_{2k})\} \end{aligned} \quad (53)$$

for station 2. At this point, both beat spectra are close to 0 Hz. As in (37), the estimate

$$\begin{aligned} \phi_k &= \left(\arg \{ \tilde{S}_{1k,beat}^{sync}(\arg \max_f \{|\tilde{S}_{1k,beat}^{sync}(f)|\}) \} \right. \\ &\quad \left. + \arg \{ \tilde{S}_{2k,beat}^{sync}(\arg \max_f \{|\tilde{S}_{2k,beat}^{sync}(f)|\}) \} \right) / 2 \\ &\approx \Theta_{1k} - \Theta_{2k} + (\gamma_1 - \gamma_2)/2 + \phi_{0k}^{sync} \end{aligned} \quad (54)$$

is comprised of the initial phase values Θ_{1k} and Θ_{2k} for each FMCW chirp and the phase shift values γ_1 and γ_2 caused by the ADC delay. Variable $\phi_{0k}^{sync} \in m\pi$ accounts for ambiguities caused by the division in (54). We multiply the beat spectra in (52) and (53) by $\exp\{-j\phi_k\}$ and phase unwrap along the chirp sequence to get the beat spectra

$$\begin{aligned} \tilde{\tilde{S}}_{1k,beat}^{sync}(f) &\approx A\delta(f - \mu\tau_0) * W(f) \\ &\quad * \mathcal{F}\{1 + j\varepsilon_k(t)\} \times \exp\{j(\gamma_1 - \gamma_2)/2\} \\ &\quad \times \exp\{j2\pi(f_c - B/2)\tau_k\} \exp\{j\phi_0^{sync}\} \end{aligned} \quad (55)$$

and

$$\begin{aligned} \tilde{\tilde{S}}_{2k,beat}^{sync}(f) &\approx A\delta(f + \mu\tau_0) * W(f) \\ &\quad * \mathcal{F}\{1 + j\varepsilon_k(t)\} \times \exp\{-j(\gamma_1 - \gamma_2)/2\} \\ &\quad \times \exp\{-j2\pi(f_c - B/2)\tau_k\} \exp\{j\phi_0^{sync}\} \end{aligned} \quad (56)$$

where ϕ_0^{sync} denotes the unknown phase of the first chirp, which cannot be corrected. The corresponding signals in the time domain are defined as $\tilde{\tilde{s}}_{1k,beat}^{sync}(t)$ and $\tilde{\tilde{s}}_{2k,beat}^{sync}(t)$. Again, intrachirp PN is modeled by linear approximation. As mentioned in Section IV-B, taking the complex conjugate of $\tilde{\tilde{s}}_{2k,beat}^{sync}(t)$ ensures that spectral peaks appear at the same position on the frequency axis.

Finally, we obtain a synthetic signal, and intrachirp PN is suppressed with the superposition of both beat spectra

$$\begin{aligned} S_{k,synth}^{sync}(f) &\approx A\delta(f - \mu\tau_0) * W(f) \exp\{j(\gamma_1 - \gamma_2)/2\} \\ &\quad \times \exp\{j2\pi(f_c - B/2)\tau_k\} \exp\{j\phi_0^{sync}\}. \end{aligned} \quad (57)$$

The signal-phase change in (57), caused by motion, is detectable, as $\gamma_1 - \gamma_2$ and ϕ_0^{sync} are constant values. However, due to the unknown ADC time delay, the initial phase cannot be recovered. Compared to the modulation scheme in Section II, we can only suppress the intrachirp PN to a limited extent depending on the quality of the timing-drift presynchronization. Therefore, phase-noise suppression by superposition increases close to the LOS signal and is weakly dependent on synchronization inaccuracy. Compared to the individual peak detection in B, the frequency-shift estimation in (51) suffers from PN. Thus, the ToF results are less accurate.

In [18], the ADC phase shift values γ_1 and γ_2 are kept constant by ramp triggering by cable to measure phase values, which are proportional to the ToF. In a fully wireless setup

with up-chirps, both the time delay and timing drift must be estimated and compensated to recover the initial phase. Alternatively, we can estimate the timing offset by presynchronizing the timing drift and using a sequence of up- and down-chirps, as shown in Section II; this is spectrally efficient too.

D. Multiplexing

More than two wireless units are required in many use cases to obtain location information in two or three spatial dimensions [2]. The local positioning systems described in [31] and [32] are based on precise presynchronization and subsequent measurement. Master-slave configurations are common as well with specific advantages and drawbacks. Fundamentally, mobile and stationary radar units operate with small individual frequency offsets. These systems simultaneously transmit, receive, and isolate all beat signals in the baseband after digital sampling. Alternatively, more than two radar units can be operated in static scenarios using time division multiple access.

The signal model described in Sections II and IV theoretically allows multiplexing by means of small offsets in frequency and/or time. Due to LO drift, efficient use of the available IF spectrum becomes rather complicated and requires a high IF bandwidth for every additional radar unit (Fig. 4). To overcome this drawback, precise presynchronization of the drift in timing is performed, as described in Section V. All wireless radar units can then be operated simultaneously in FD mode and spectra can be allocated efficiently.

Initially, all radar units are coarsely synchronized in time and frequency, e.g., by the method described in [21]. The sampling criterion is thus fulfilled. Thereafter, an arbitrary radar unit is chosen as reference. The LO drift of all other stations must be estimated and corrected with the method in Section IV. Now, the drift in timing is compensated and all timing and frequency offsets are known. For simultaneous measurement with all radar units, the IF signals are separated by utilizing small offsets in time and/or frequency. See [33] for requirements details. The measurement chirp sequence is comprised of either up- or down-chirps or interleaved up- and down-chirps. First, the down-converted signals are sampled digitally; then, data from all radar units are transferred to a CPU or to a single radar unit.

As the unknown parameters can be determined precisely from the ADC data from all radar units, rough synchronization in time and frequency is only required to meet the sampling criterion during the first measurement sequence. The main advantages of this multiplexing scheme are: 1) all radar units can scan at the same time; 2) the synchronization errors are suppressed; 3) PN is reduced; and 4) the velocity vector can be measured by geometric considerations. A 2-D or 3-D displacement can be detected as well by coherently processing all signal paths. The challenge with this sort of setup is to have a sufficiently high transmission rate to transfer sampled data from all wireless radar units.

VI. SIMULATION

To quantify the effect of PN suppression by superposition and to obtain the theoretical limit of the approximation in (36),

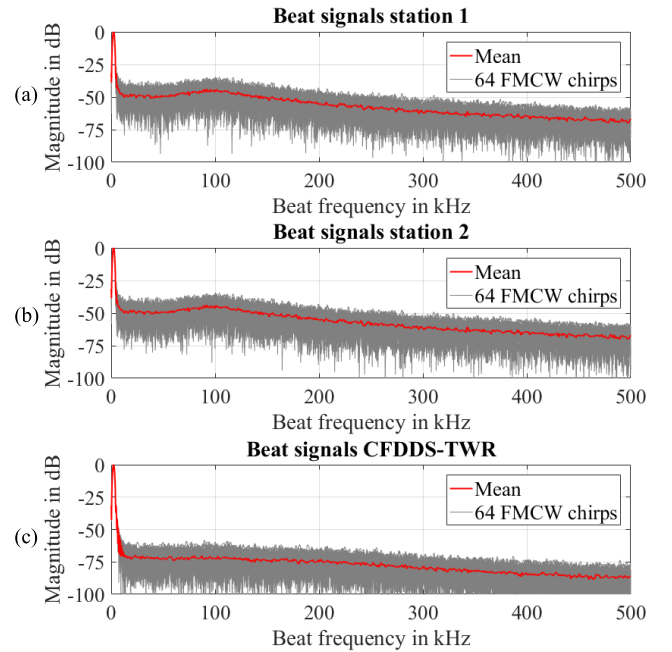


Fig. 7. Simulation of the beat signals from (a) station 1, (b) station 2, and (c) superposition of both spectra with CFDDs-TWR. Sixty-four individual FMCW chirps and the mean value are depicted in gray and red, respectively. All signals are normalized to a peak value of 0 dB and the LOS length is $x_0 = 2.75$ m. The superposition improves the SNR by 27 dB at a beat frequency of $f = 1$ kHz.

we explain a simulation of this operation in 1-D and 2-D. We choose the radar parameters (summarized in Table II) to suit the available test hardware. Carrier frequency, bandwidth, and sweep time are set to $f_c = 24$ GHz, $B = 250$ MHz, and $T_{sw} = 1$ ms, respectively. The length of the chirp sequence, comprised of up- and down-chirps, is $K = 64$ and the simulated one-way range in this static scenario is $x_0 = 2.75$ m. We measured the hardware PN profile and used it here for accurate comparison purposes. The effects of PN are much more severe than those of thermal noise, especially in indoor scenarios. There is no PN suppression due to correlation, as with primary radar systems. Therefore, we set the thermal noise floor for this simulation to -80 dB, which corresponds to the dynamic range of a 14-bit ADC. To obtain the upper bound on the achievable suppression of PN caused by incoherent LOs, both wireless units are assumed to be perfectly synchronized in time and frequency.

Fig. 7 depicts the beat signals for station 1 (a), station 2 (b), and the superposition of these signals with CFDDs-TWR (c). The beat signals from 64 FMCW chirps are colored gray and the mean is shown in red like the persistence mode in an analog oscilloscope. Fig. 7 shows a reduction of the PN floor by 27 dB at a beat frequency of $f = 1$ kHz for the superposition (at a PN level of -71 dB) with respect to the individual signals (at a PN level of -44 dB). The 2-D FFT of the chirp sequence is depicted in Fig. 8, where the spectrum on station 1 (a) and station 2 (b) are superimposed with CFDDs-TWR (c). The cross section at a target range of $x_0 = 2.75$ m is shown in the lower portion of Fig. 8. The improvement is clearly visible as the shape of the 2-D Hanning window appears with an increased number of sidelobes. Except at the LOS range,

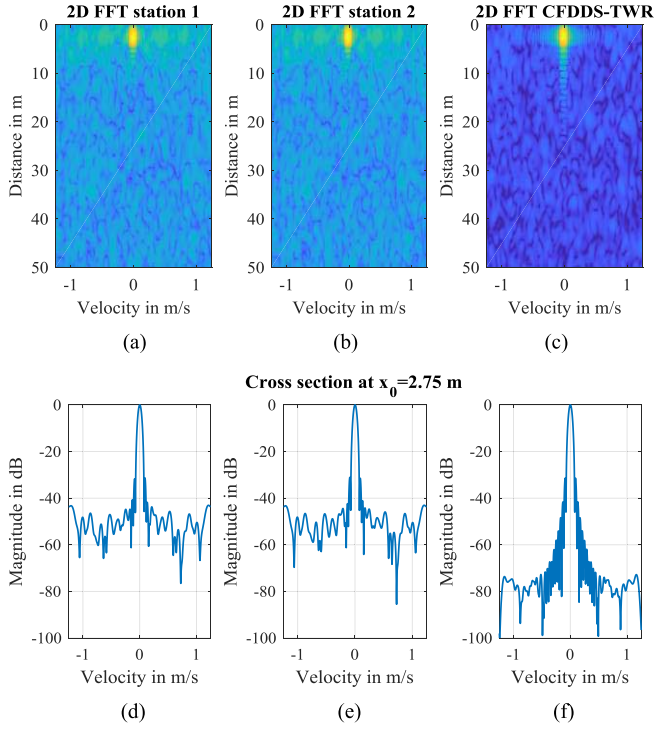


Fig. 8. Simulation results for a 2-D FFT of the beat signals with 64 FMCW chirps at (a) station 1, (b) station 2, and (c) superposition of both signals with CFDDS-TWR. The LOS distance was $x_0 = 2.75$ m, the velocity $v = 0$ m/s and all signals are normalized to peak value. Deep blue and bright yellow colors indicate a magnitude of -100 and 0 dB, respectively. The noise level at stations 1 and 2 was at -69 dB and the superposition results were in a noise floor of -83 dB. The corresponding cross sections at an LOS range of $x_0 = 2.75$ m are depicted in (d)–(f).

this operation reduces the noise floor of the range-Doppler image from -69 dB to -83 dB. At x_0 , the mean PN level close to the LOS is substantially higher (-53 dB) as the signal amplitude is corrupted by different realizations of the PN across the slow-time dimension [(d)–(f)]. Hence, fluctuations of the amplitude of individual chirps are mapped onto the entire unambiguous velocity by an FFT in this dimension. The mean PN level at the distance x_0 is also reduced to -83 dB by means of the superposition.

VII. MEASUREMENTS

First, we shall review the test hardware. The ranging tests are carried out with one radar system mounted on a linear unit and one stationary radar in a laboratory environment (Fig. 9). Tests with strong LOS and weak multipath propagation will be formulated later. We will compare the superposition with the simulation results in Section VII-A to demonstrate the effectiveness of the PN suppression by the proposed algorithms with experimental data. We will then show how range and velocity estimation were improved with the test results. Finally, we will review the practical performance of phase tracking for displacement measurements.

A. Overview of the Measurement Hardware

The block diagram of the test hardware is given in Fig. 10. Both radar units have identical hardware. The main component is the transceiver monolithic microwave integrated circuit (MMIC) BGT24MTR12, which is controlled by a CPU

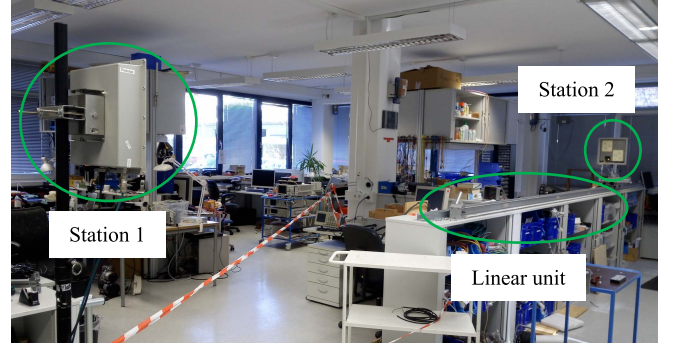


Fig. 9. Measurement setup with a linear unit and two radar stations. Range and velocity measurements are taken with strong LOS and weak multipath.

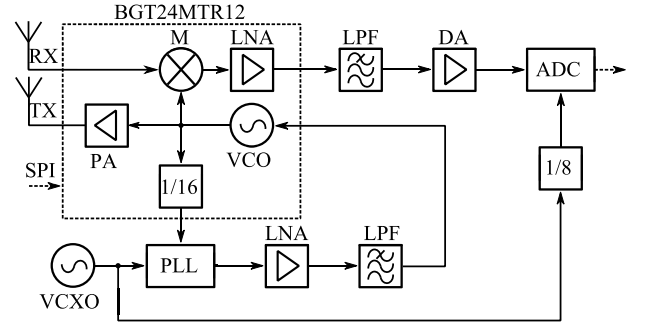


Fig. 10. Block diagram of relevant components of the 24-GHz FMCW radar system.

(Xilinx Zynq XC7Z020) via serial peripheral interface. Internally, it is comprised of a power amplifier, an RF mixer (M), a low-noise amplifier (LNA), a voltage-controlled oscillator (VCO) and a $1/16$ clock divider. The VCO is stabilized by an external type ADF4159 fractional- N phase-locked loop (PLL) with subsequent LNA (TS9711LT) and low pass loop filter (LPF). TX signals are generated, and RX signals are received by this MMIC which is connected to two patch antennas. The down-converted signals of the MMIC are filtered by an LPF and further amplified with a type AD8132 differential amplifier. The type Y700 VCXO serves as reference clock for the PLL and the CPU and supplies the ADC (LTM9006) after division by a factor of 8. Therefore, the sampling frequency in baseband is 16 MHz.

B. Multipath Propagation

The signal models above are derived assuming LOS operation and no thermal noise. Multipath propagation impacts the performance of these systems, especially in indoor localization [34]. To cater for multipath, we adapt the model of the synthetic beat signal in (40) and (57) to

$$S_{k,\text{synth}}^{u/d}(f) = \sum_{p=1}^P \{A_p \delta\{f - \mu\tau_{0,p}\} * W(f) \exp\{j\phi_0^{u/d}\} \cdot \exp[\pm j2\pi(f_c \mp B/2)\tau_{k,p}]\} + N(f) \quad (58)$$

where p and $N(f)$ denote the p th signal path and the Fourier transform of AWGN, respectively. In total, signal propagation can occur along P paths with different ToF values. The method described above requires a dominant LOS signal to estimate unknown system parameters. In strong multipath environments,

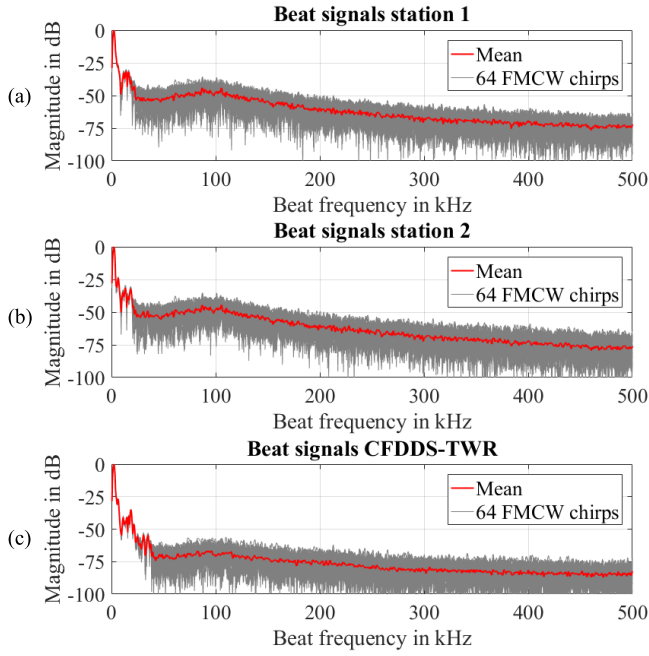


Fig. 11. Measurement of the beat signals of (a) station 1, (b) station 2, and (c) superposition of both spectra with CFDDS-TWR. Sixty-four individual FMCW chirps and the mean value are depicted in gray and red, respectively. All signals are normalized to a peak value of 0 dB and show weak multipath propagation close to the LOS with a length of $x_0 = 2.75$ m. The superposition results in an improvement of 24 dB at $f = 1$ kHz.

a common antenna for TX and RX can be used and the maximum detection in (26) or (51) has to be replaced by a convolution of the beat signal in (25), (46), and (47), respectively, and subsequent detection of the maxima. In addition, the phase correction values in (37) and (39) or (50) and (54) can be determined by multiplication of the beat spectra after the detection and correction of the frequency shift [see (51)]. This allows for the estimation of the unknown offsets independent of the channel impulse response and suppression of stationary clutter in non-static scenarios. Furthermore, coherent ranging with CFDDS-TWR enables multipath suppression with an inverse SAR approach [35]. To obtain accurate ranging results in static scenarios, the RF bandwidth must be large enough to separate the LOS, which is the shortest path, from other components in the frequency domain.

C. PN Suppression Measurement

All measurements were carried out with one stationary radar unit as shown in Fig. 9. The second radar station was mounted on a linear unit which serves as precise reference. A strong LOS was available throughout all tests. Weak multipath propagation, caused by reflections at the linear unit, the ceiling and other objects, is expected to have a minor effect on the results. The first measurement with interleaved up- and down-chirps was conducted in a static scenario to gain insight into the PN-suppression results. All parameters (Table II) are the same as the simulation parameters mentioned in Section VI. Presynchronization ensures that the requirements in Table I are met. Due to the short ToF or range, thermal noise was smaller than the quantization noise of the 14-bit ADC with a dynamic range of 80 dB. As depicted in Fig. 11, the measured PN

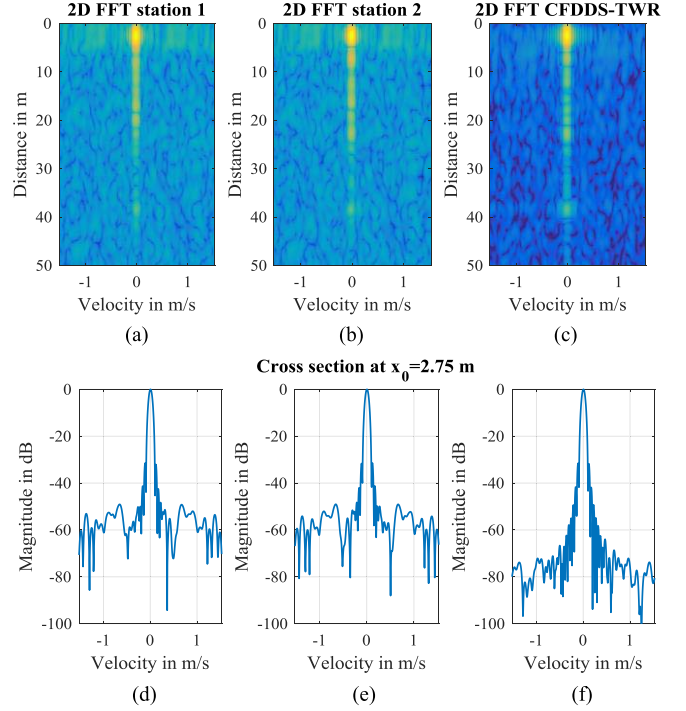


Fig. 12. Measurement results with the 2-D FFT of the beat signals with 64 FMCW chirps of (a) station 1, (b) station 2, and (c) superposition with CFDDS-TWR for an LOS distance of $x_0 = 2.75$ m and a velocity of $v = 0$ m/s. All signals are normalized to the peak value. The color scaling is the same as in Fig. 7. The individual noise floor was -67 dB, which was reduced to -83 dB by superposition. The corresponding sectional view at the LOS distance $x_0 = 2.75$ m is shown below the 2-D FFTs in (d)–(f).

level of approximately -44 dB at $f = 1$ kHz is the dominant influence on the radar performance [(a) and (b)]. The reduction of 24 dB to a noise level of -68 dB with CFDDS-TWR (c) at this frequency is comparable to the simulation results as it differs only by 3 dB. Weak multipath propagation is visible at distances above $x_0 = 2.75$ m. The 2-D Fourier transform is depicted in Fig. 12 for the beat signals at station 1 (a), station 2 (b), and the superposition of these signals (right). The noise floor was reduced to -83 dB by superposition as against a noise floor of -69 dB for each individual station. This is the same as the results in Section VI. Again, the cross sections at a $x_0 = 2.75$ m range are depicted in (d)–(f) below the 2-D FFTs; they demonstrate a mean reduction by superposition from -55 dB to approximately -80 dB. The PN levels and the reduction are also comparable to the simulation results.

D. Range and Velocity Measurements

To demonstrate the improved range and velocity estimates, we measure the standard deviation (SD) of our algorithms in a static scenario with an accurate reference. While all radar parameters are the same as in the measurement setup in C, the bandwidth was increased to $B = 500$ MHz. The range resolution is $\delta_x = 2c_0/B = 1.2$ m, where the factor 2 accounts for the influence of the Hanning window function. One radar unit was moved from 6.9 to 16.9 m in uniform steps of 0.2 m. Sixty-four FMCW bursts with $K = 64$ up-chirps are evaluated at each position with algorithms that rely on precise presynchronization, as described in Section V.

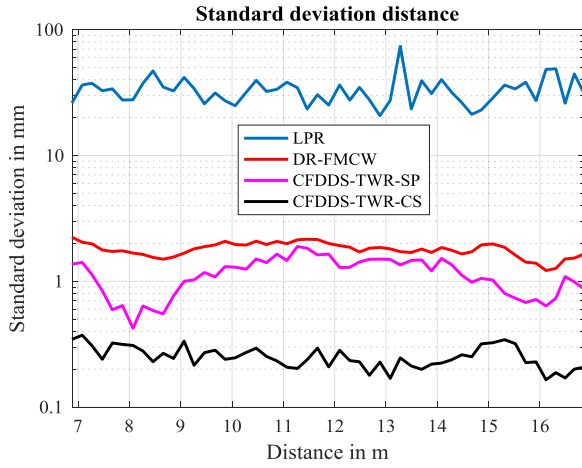


Fig. 13. SD of the range estimate in a static scenario with weak multipath. Measurements were conducted at LOS distances between 6.9 and 16.9 m at 51 positions with a uniform step size of 0.2 m and 64 repetitions. The mean SD was 3.3 cm for the LPR in [6] (blue), 1.8 mm with the DR-FMCW in [17] (red), 1.1 mm with superposition (CFDDS-TWR-SP, magenta) and 0.25 mm with chirp sequence secondary radar with (48) and (49) (CFDDS-TWR-CS, black).

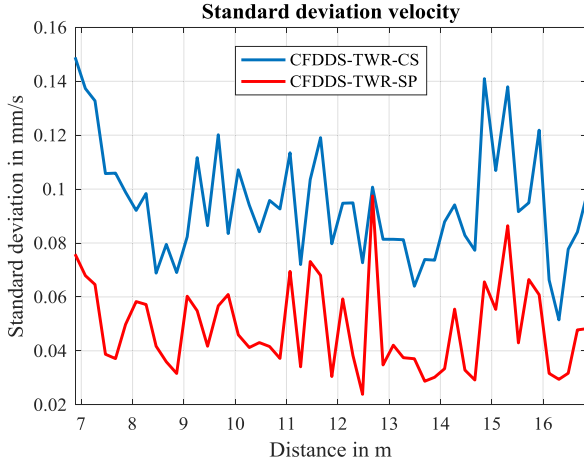


Fig. 14. SD of the velocity estimation with CFDDS-TWR in a static scenario. The mean SD was 94 $\mu\text{m/s}$ for the chirp sequence phase estimation with (50) (CFDDS-TWR-CS, blue) and 48 $\mu\text{m/s}$ for the 2-D FFT of the superposition (CFDDS-TWR-SP, red).

After each up-chirp, we use the time T_{sw} to reset the PLL. Therefore, the measurement duration is $128T_{\text{sw}} = 128$ ms and the processing time with MATLAB 2018b, a 3.1-GHz Quad-Core CPU, and 16-GB RAM is approximately 0.5 s. The time between consecutive bursts, i.e., the refresh time including data transmission, is 1 s. Fig. 13 shows the SD of the range estimation at each position. The results of the CFDDS-TWR via superposition in (57) are slightly worse (1.1 mm instead of 0.25 mm) than the chirp sequence estimation with (49) as the frequency-shift estimation in (51) suffers from PN.

The results of the velocity estimation with CFDDS-TWR in this static scenario via (57) and individual chirp sequence evaluation, according to (50), are depicted in Fig. 14 in terms of the SD at every position. Again, a Hanning window is used in slow time dimension which yields a velocity resolution of $\delta_v = c_0/(f_c T_0 K) = 0.2$ m/s. The variable $T_0 = 2T_{\text{sw}}$ defines the time between two consecutive FMCW chirps. Theoretically, the SD of the coherent superposition with

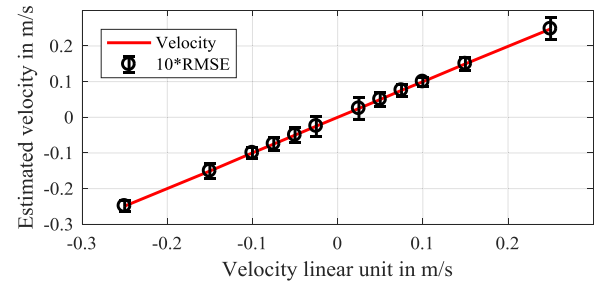


Fig. 15. LOS velocity measurement with a linear unit and reference velocities of $v_{\text{ref}} = \pm\{25; 50; 75; 100; 125; 250\}$ mm/s with CFDDS-TWR. Ten RMSEs are depicted in black as confidence intervals for every reference velocity. The average RMSE was 2.1 mm/s, which is higher than the SD of the static radar measurement.

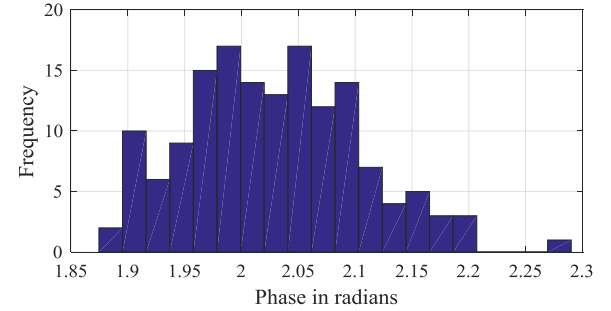


Fig. 16. Histogram of the phase tracking results with strong LOS at an average range of $x_0 = 2.7$ m. The LOS distance was increased with 153 steps and a displacement of $\Delta x = 2$ mm. The measured phase shift mean per step was $\Delta\phi_0^u = 2.03$ rad and the resulting SD of the displacement was 75 μm for an RF carrier of $f_c = 24$ GHz.

AWGN is 3 dB lower than the individual peak detection. Compared to the measured SD of the wireless positioning system in [6], which is stated in [36] to be 0.16 m/s, the results of coherent chirp sequence processing are greatly enhanced to 94 and 48 $\mu\text{m/s}$.

To estimate the root-mean-square error (RMSE) of the velocity estimation, the linear unit provides reference velocities of $v_{\text{ref}} = \pm\{25; 50; 75; 100; 125; 250\}$ mm/s. The radar unit thus moved from $x_0 = 6.9$ m to $x_0 = 16.9$ m at constant velocity. The measured RMSE values multiplied by 10 are depicted with black error bars in Fig. 15 for every reference velocity; the red solid line traces the measured velocity against the reference. The estimated mean radar velocity differs by only 11 $\mu\text{m/s}$ from the reference velocity. The average RMSE here was 2.1 mm/s, which is clearly worse than the SD in the static scenario in Fig. 14. Hence, we expect the limited accuracy of the linear unit to be the main error source in this trial.

E. Phase Tracking

Finally, we assess the performance of the displacement measurement with CFDDS-TWR and phase tracking. To this end, the initial one-way distance of 2.547 m was increased to 2.851 m by 153 steps of $\Delta x = 2$ mm and the phase difference of two consecutive measurements was evaluated via (39) at all positions. For this radar system with a carrier frequency of $f_c = 24$ GHz, we expect a $\Delta\phi_0^u = 2.01$ phase shift for the selected step size with respect to the round trip ToF. A statistical evaluation of the data in the histogram in Fig. 16

TABLE III
COMPARISON OF RADAR LOCALIZATION METHODS

Reference	Ranging SD	Velocity estimation SD	Phase tracking SD	Method
[6]	1 cm	—	—	LPR
[7]	±10 cm	—	—	2D TDoA
[9]	6.2 m	—	—	RSS
[16]	0.1–0.3 m	—	—	OTAD
[17]	0.21 mm	—	—	DR-FMCW, cable
[18]	0.5 mm	—	—	DR-FMCW, RX array
[21]	1 cm	—	—	LPR, cable
[23]	1 cm	—	—	LPR
[27]	<12 mm	—	—	DR-FMCW
[32]	2.7 cm	—	—	LPR
[36]	6.9 mm	0.16 m/s	—	LPR
This work	0.25 mm	48 µm/s	75 µm	CFDDS-TWR

Comparison with the state of the art.

yields a measured mean phase shift of $\Delta\hat{\phi}_0^u = 2.03$ with a SD of the displacement of 75 µm. Therefore, the measurement results are very close to the expected values.

F. Comparison of Localization Methods

A comparison of selected radar localization methods in terms of the SD is given in Table III. Imprecise ranging can be carried out with the RSS and OTAD principle. The ranging precision of the LPR method is strongly limited to centimeters as the measurement suffers from PN and requires precise presynchronization. These limitations can be overcome with the DR-FMCW allowing a SD of 0.21 mm which was demonstrated in a measurement setup with cable in the 5.8 GHz band. The ranging SD of CFDDS-TWR and DR-FMCW with a wireless setup in this paper were 0.25 and 1.8 mm, respectively. Velocity measurements are enabled by the LPR principle via beat frequency evaluation with a SD of 0.16 m/s. An improvement by a factor of 3300 with the phase based CFDDS-TWR has been demonstrated in this paper. So far, displacement measurements with phase tracking and separate wireless radar stations have not been reported by other authors. Here, a phase tracking SD of 75 µm has been measured.

VIII. CONCLUSION

In this paper, we upgraded wireless positioning systems based on FMCW secondary radar to chirp-sequence-based systems. As the localization performance of separate radar units is mainly impacted by LO drift and PN, we presented two different CFDDS-TWR implementations. In one, all unknown parameters and the initial phase can be estimated with interleaved up- and down-chirps. In the other, precise presynchronization of the LO timing drift eases the sampling requirements, enables a doubling of the unambiguous velocity and facilitates multiplexing. In both CFDDS-TWR configurations, range and velocity are estimated with computationally

efficient 2-D FFTs of the synthetic beat signal. The measured PN suppression of this synthetic 24 dB beat signal was close to the simulation result of 27 dB. We investigated the effects of the enhanced dynamic range on the accuracy of the range and velocity measurements too. We also demonstrated a SD of displacement measurement results of 75 µm with continuous phase tracking.

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